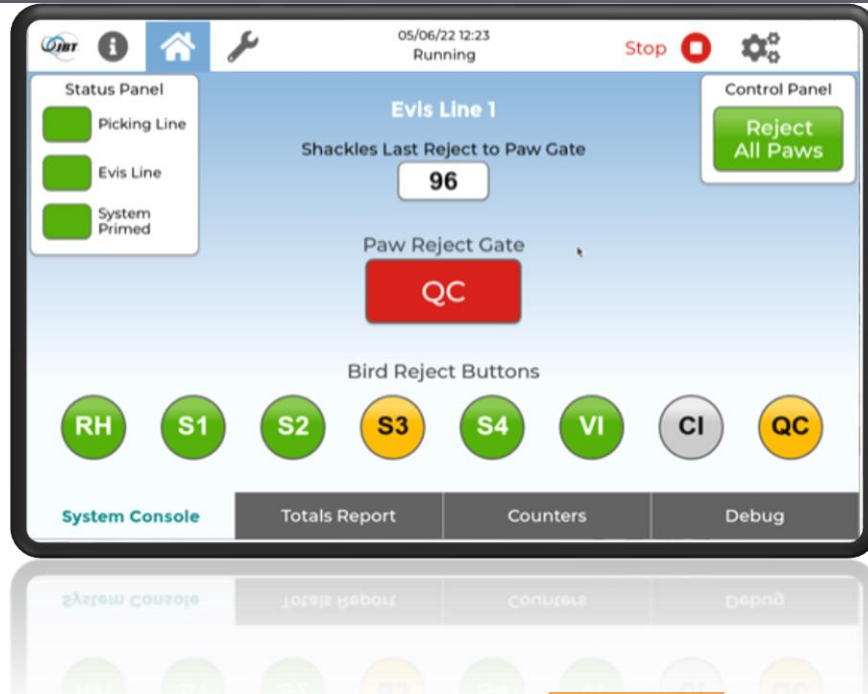




**MARS**  
FOOD PROCESSING SOLUTIONS



## PAW TRACKING SYSTEM



Customer:

Sales Order #:

Serial #:

### Safety Instructions & Technical Manual

Prepared by:

Engineer Approval:

Approved by:

|   |                                    |
|---|------------------------------------|
| 1 | General Information                |
| 2 | Operations and Instructions        |
| 3 | Sanitation                         |
| 4 | Maintenance and Troubleshooting    |
| 5 | Assembly Drawings                  |
| 6 | Controls and Electrical Schematics |
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## GENERAL INFORMATION

### SAFETY GUIDELINES



#### **DANGER!**

#### **WARNING!**

All operations, maintenance and sanitation personnel **MUST** read and understand this manual and become familiar with the equipment before operating, maintaining or sanitizing.

Operators **MUST REMOVE** all **RINGS WATCHES NECKTIES, LOOSE GARMENTS**, and any other items that might become entangled in the moving parts of the machine. **NO CHAIN MESH GLOVES** should be worn while operating this machine.

Operators **MUST KEEP HANDS AWAY** from the **CUTTING AREA** and any **ROTATING or MOVING PARTS** at all times.

This machine is equipped with operating safety interlocks, which must be fully functional at all times. No attempt should ever be made to bypass safeties or run the machine with any guard removed.

Do not attempt to service machine unless power has been disconnected and locked out/ tagged out, and the machine has come to a complete stop.

NOTE: Exceptions to OSHA 29 CFR 1910.147 – Control of Hazardous Energy (Lockout/Tagout Standard) are as follows below.

Minor tool changes, adjustments and other minor service activities that take place during normal production are not included in the Lockout/ Tag out Standard provided they are:

- Routine, repetitive and integral to the use of the equipment
- Performed using alternative measures that provide effective protection for the employee



**Failure to heed these warnings and comply with the instructions for safe operation of this machine can and will cause serious bodily injury and or death.**

## GENERAL INFORMATION

### WARNING LABELS AND LOCATIONS



**SAFETY STICKER MAY COME OFF DUE TO USE OF HIGH-PRESSURE WATER. IF A STICKER IS MISSING IT MUST BE REPLACED IMMEDIATELY.**

| PART NO. | DESCRIPTION                               | QTY. | IMAGE                                                                               |
|----------|-------------------------------------------|------|-------------------------------------------------------------------------------------|
| 19883    | SAFETY STICKER, WARN, VOLTAGE, HORIZONTAL | 1    |  |

## System Description

**JBT Paw Tracking System** maintains precise track of chicken paws after they are separated from the rest of the bird, so that the paws can be automatically discarded when a USDA official rejects a chicken on the evisceration line. The paws are tracked using a series of photoeyes on the Kill Line and the Evisceration Line, and by knowing how many shackles are between the various photoeyes and the USDA Inspection Stations. If the system uses a Paw Conveyor at the end of the Kill Line, Timed Virtual Shackles keep track of the paws as they travel the Paw Conveyor to the Paw Reject Gate.

The words "photoeye" and "sensor" are used interchangeably in this manual.

| Options                                               |
|-------------------------------------------------------|
| Kill line that splits to single evis, or 2 evis lines |
| Up to 12 Reject Buttons on the evis line              |
| 2 to 4 brackets on the evis line                      |
| Fixed speed or variable speed conveyors               |

### JBT Efficiency Management System

Some installations also include JBT EMS (Efficiency Management System) software that runs on a Windows computer in a nearby office. The JBT EMS software continuously uploads data during the day and stores it in a Microsoft SQL Server database. The program makes this data available to users via reports, charts, status screens, printing, automatic report emailing, alarms and many other useful features. There is also a client version of the software that allows users to run the program on their own computer in their office.

## GENERAL INFORMATION

### EQUIPMENT TRAINING RECORD

All Sanitation and Maintenance personnel must be trained in the following areas. Prime Equipment Group recommends you keep a record of personnel training. Prime Equipment Group is willing to assist with training, following a request to do so. Please feel free to call Prime Equipment Group for additional information.

- Installation and product application
- Disassembly and sanitation
- Instruction and parts manual
- Daily periodic maintenance
- Wearing parts
- Assembly and setup for operation
- Safety features and precautions
- Troubleshooting
- Recommended spare parts inventory
- Routine preventative maintenance schedule



## GENERAL INFORMATION

### LIMITED WARRANTY

Products purchased from Manufacturer are warranted only by the Manufacturer's warranty that the Product provided herein is of their standard quality and operable and that the Product will be free from defects in workmanship and material. The warranty period, for Product sold by Manufacturer and not expressly made subject to a different warranty, is one hundred twenty (120) days from date of receipt by Customer. There is no insurance against or warranty of any type for production loss due to late or damaged Product shipments. The Manufacturer makes no express or implied statutory warranties other than as expressly set forth herein.

Manufacturer shall for a period of one hundred twenty (120) days from the date any Product sold hereunder is received by Customer, repair or replace at its option, free of charge and F.O.B. point of manufacture, any nonconforming or defective parts or Product that upon inspection by Manufacturer, are deemed to be nonconforming or defective. Manufacturer shall, however, only be required to repair or replace such parts or Products if Customer (i) owned the nonconforming or defective parts or Product continuously from the original date of delivery, (ii) stored and maintained such parts and Product in accordance with Manufacturer's recommendations and standard industry practices and (iii) notified Manufacturer in writing of any nonconformance or defective parts or Product within fifteen (15) days following the date the nonconformity or defect became reasonably apparent. Further the obligations of Manufacturer hereunder do not apply to the repair or replacement of Products within the aforementioned warranty period for (a) misuse, negligence or accident, (b) normal maintenance services, (c) normal replacement of service items in connection with such service, (d) normal deterioration due to wear and tear, (e) use of non-manufacturer (OEM) replacement parts. The remedy provided in this Warranty shall be the sole and exclusive remedy of Customer in the event of defective or nonconforming parts or Product, and shall be void if the Product has been altered in any manner or has been moved from its original location without the written approval of the Manufacturer. The repair or replacement of any parts or Products

under the foregoing provision does not extend the warranty beyond the warranty period described above. All electric motors, controls and other electrical services and their component parts, hydraulic motors, controls and other hydraulic services and their component parts, pneumatic controls, pneumatic services and other pneumatic components, bearings, gearboxes etc. are not included within this Warranty the warranty with respect to these parts being limited to the manufacturer of these parts, a copy of which may be obtained from Manufacturer upon request. Customer will be responsible for the shipping/packaging costs associated with the safe and expedient return of warranted parts. The Warranty can only be claimed if genuine Manufacturer spare parts are used for repairs and maintenance of the Product.

**MANUFACTURER'S UNDERTAKINGS AS CONTAINED IN THIS WARRANTY AS SPECIFIED ABOVE SHALL BE IN LIEU OF ANY OTHER GUARANTY OR WARRANTY EXPRESS OR IMPLIED INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTY OF MERCHANTABILITY OR IMPLIED WARRANTY OF FITNESS FOR A PARTICULAR PURPOSE. IN ADDITION IT IS EXPRESSLY AGREED BETWEEN THE PARTIES THAT THE MANUFACTURER HAS NOT MADE ANY REPRESENTATIONS OR WARRANTY AND THERE IS NO AGREEMENT THAT THE PRODUCT PURCHASED HEREUNDER WILL MEET ANY PARTICULAR**

**STANDARD UNLESS CUSTOMER HAS SUPPLIED MANUFACTURER WITH WRITTEN SPECIFICATIONS OF THE PRODUCT AND THE MANUFACTURER HAS AGREED IN WRITING THAT THE PRODUCT WILL MEET SPECIFIC STANDARDS SPECIFIED BY THE CUSTOMER. CUSTOMER UNDERSTANDS AND ACKNOWLEDGES THAT THERE ARE NO WARRANTIES WHICH EXTEND BEYOND THE TERMS HEREOF UNLESS OTHERWISE NOTED IN PURCHASE AGREEMENT.**

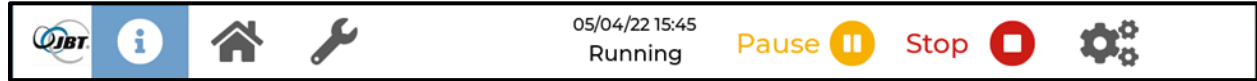
**NO CLAIM OF ANY KIND INCLUDING BUT NOT LIMITED TO CLAIMS OF NEGLIGENCE AND BREACH OF CONTRACT WHETHER AS TO PRODUCT DELIVERED OR SERVICES RENDERED OR FOR NON-DELIVERY OF PRODUCT OR NON-PERFORMANCE OF SERVICES SHALL BE GREATER THAN THE PURCHASE PRICE OF THE PRODUCT OR SERVICES IN RESPECT OF WHICH THE CLAIM IS MADE. WITHOUT LIMITING THE FOREGOING MANUFACTURER'S LIABILITY HEREUNDER SHALL UNDER NO CIRCUMSTANCES EXCEED THE PURCHASE PRICE SET FORTH IN THIS QUOTATION.**

**MANUFACTURER SHALL IN NO EVENT BE LIABLE FOR ANY INCIDENTAL CONSEQUENTIAL INDIRECT OR SPECIAL DAMAGES LOSS OF BUSINESS PROFIT PERSONAL INJURY BUSINESS INTERRUPTION AND LOSS OF BUSINESS EVEN IF MANUFACTURER HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES AND ANY CLAIM BY CUSTOMER TO SUCH DAMAGES WHETHER BASED ON TORT CONTRACT OR OTHERWISE IS HEREBY EXPRESSLY WAIVED AND EXCLUDED.**

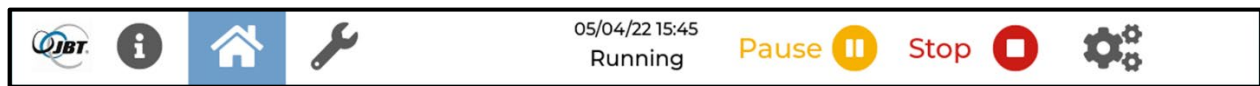
**OPERATIONS AND INSTRUCTIONS**

**SPECIFICATIONS**

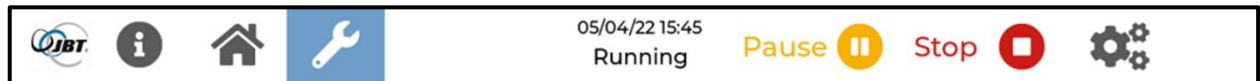
### JBT System Navigation Site Map



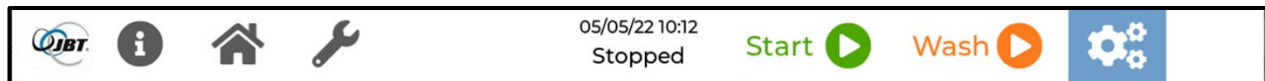
- **Info/About**
  - Company
  - Contact Info
  - Software Version



- **Home**
  - **System Console**
    - Counter
    - Debug
      - Show Block Times, Compare Sensors, Trolley Debug Data



- **Tools**
  - Operator
    - Reject Buttons
    - Line Count Data
  - Maintenance
    - Reject Settings
    - Blocktimes
    - Shackle Counts
    - Conveyors/Gates
    - Line Speeds
    -
  - Hardware
    - Hardware Test Screens
    - I/O List

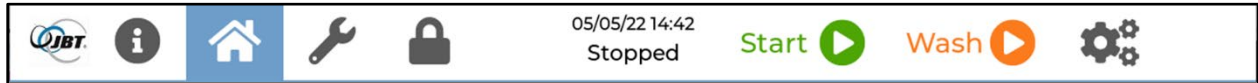


- **Advanced Settings**
  - **Advanced**
    - Controller
    - Security
  - **Admin**
    - General
    - JBT Adv Admin

## OPERATIONS AND INSTRUCTIONS

### NAVIGATION BAR

## Navigation Bar



The Navigation Bar is the 1<sup>st</sup> layer of navigation. The selected icon is shown in blue.

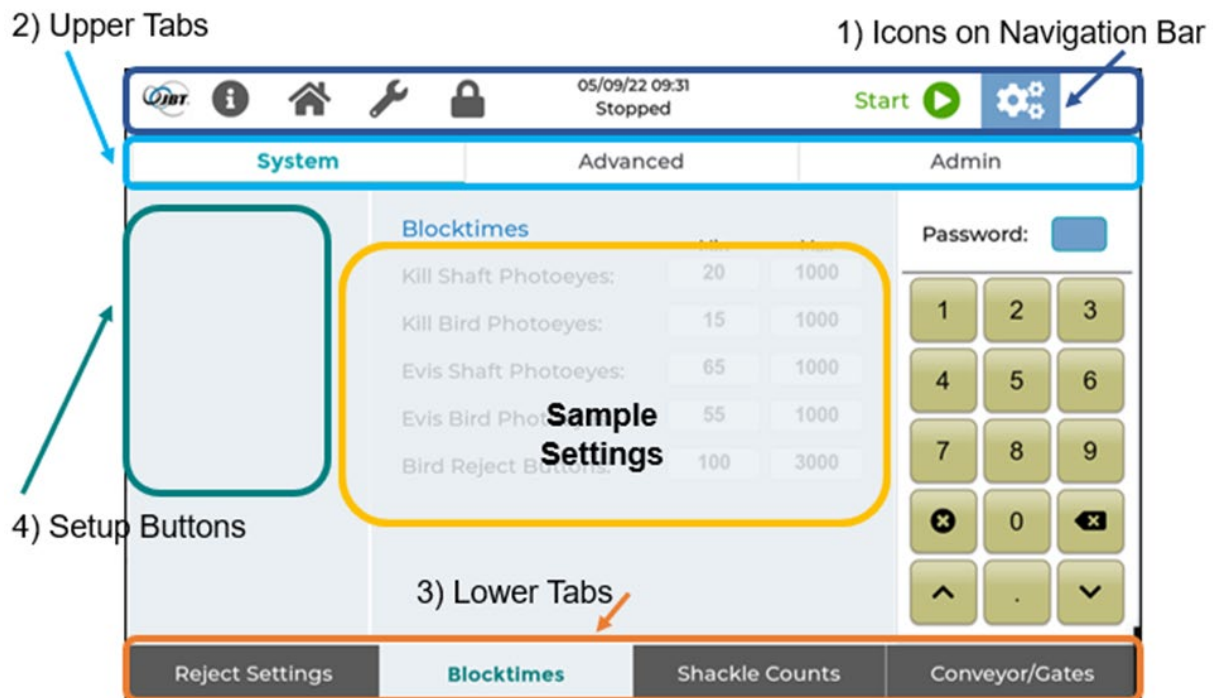
| Icon                      | Name                                                             | Description                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | Info (About)                                                     | Info about the company, contact info, program name and version.                                                                                                                                                                  |
|                           | Home                                                             | The home screen shows the system in action. Startup and time-outs default to this screen.                                                                                                                                        |
|                           | <b>Operator System</b>                                           | The Tools section contains tools used by operators and their supervisors.                                                                                                                                                        |
|                           | Lock Screen                                                      | Lock the system (optional icon)                                                                                                                                                                                                  |
|                           | Logged on Security Level<br><br><b>Touch to manually logout.</b> | <ul style="list-style-type: none"> <li>Operator (Green): not shown if operator passwords are turned off.</li> <li>Maintenance (Purple)</li> <li>Advanced (Orange)</li> <li>Super User (Red) Used during installation.</li> </ul> |
| 05/04/22 15:45<br>Running | System State                                                     | The center of the navigation bar shows the current time and system state                                                                                                                                                         |
| Start                     | <b>Start</b>                                                     | Start the System. There is an option on Controller Setup to Autostart after Power-Up.                                                                                                                                            |
| Stop                      | <b>Stop</b>                                                      | Stop the System. Passcord required to stop the system.                                                                                                                                                                           |
|                           | <b>Advanced Settings</b>                                         | The Settings section contains advanced System Settings, Security Settings, and Admin Settings.                                                                                                                                   |

### JBT System Navigation

#### Navigation Overview

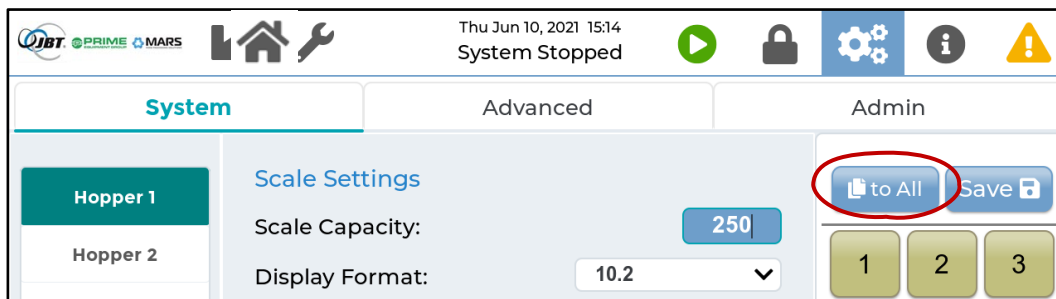
Navigation occurs in the following steps. Some screens may not use each of these steps (for example, the screen may not need setup buttons on the left), however the steps are always in this order.

Sample navigation steps to a setup screen:



**Settings** (Cogs) – **System** (Upper Tab) – Block Times (Lower Tab) – **No Setup Buttons**

**Copy to All** button copies the values of the 1<sup>st</sup> entity to other entities.

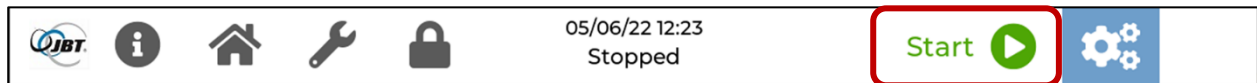


## OPERATIONS AND INSTRUCTIONS

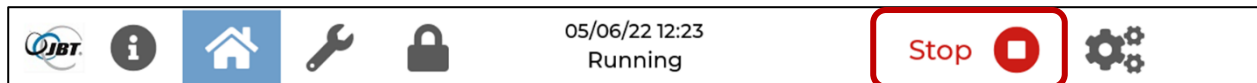
### SYSTEM CONTROLS

## System Console – System Controls

System controls are located on the navigation bar.



The **Paw Tracking System** is started by touching the **Start** button on the **Navigation Bar**. It is also started automatically if a button isn't touched within 20 seconds after powering up the **JBT Controller**. An option to enable/disable autostart on power-up is on Settings (Cogs) – Advanced (Top Tab) – Controller (Bottom Tab)



Touch the Stop icon to stop the system. A Level 1 Operator or above password is required to stop the system.

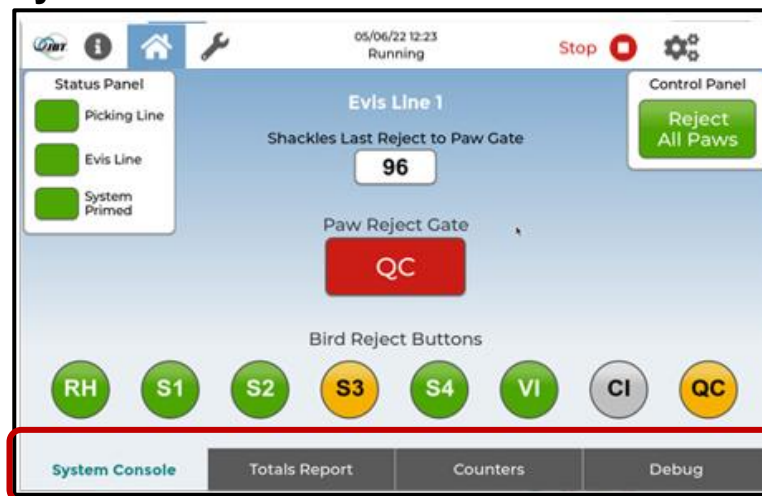
## Adjusting Settings

Touch the **Tools (Wrench)** icon to access the **Operator Tools** such as Bird Reject Button activation, Hardware Test, or Line Count Setup.

Touch the **Settings (Cogs)** icon for system configuration settings..

**It is not necessary to stop the system to adjust the settings.**

## System Console Screens



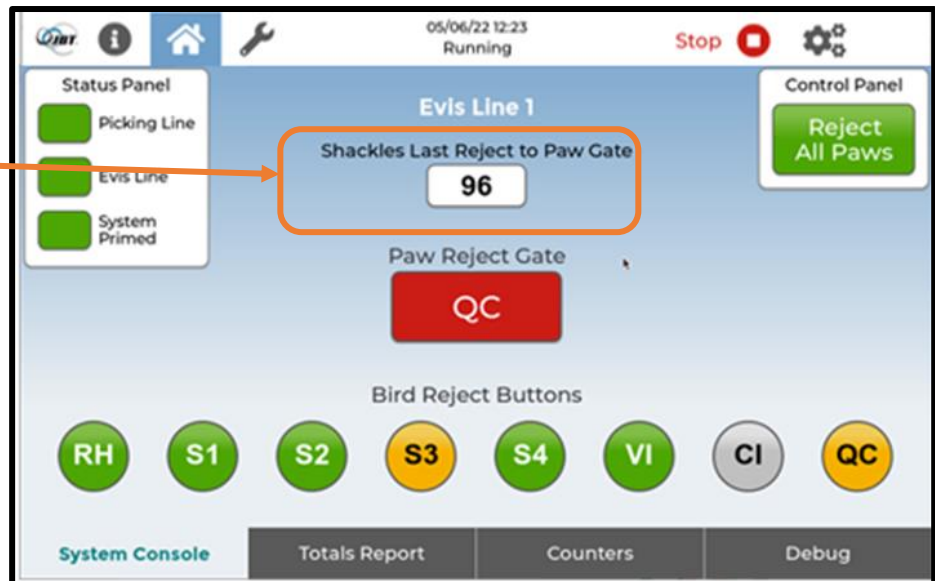
Screens associated with the system console/home screen are accessed by the tabs across the bottom of the home screen

### System Console – Paw Screen

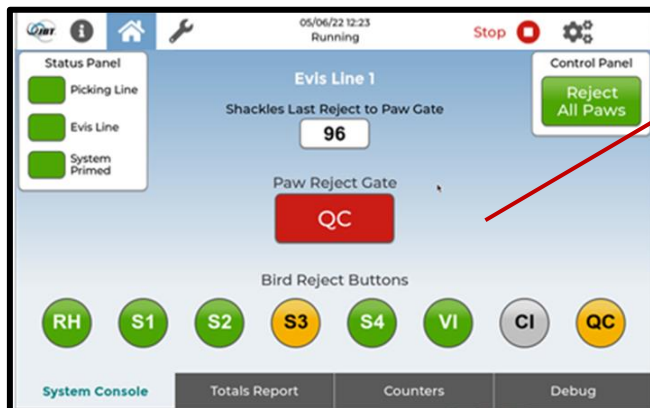
The System Console shows the current status of your system. The colored blocks on the System Console show the status of the indicated line(s), gate(s), and Bird Reject Buttons. Your System Console screen will display your plant's configuration of USDA reject buttons and paw reject gates.

**Shackles Before Last Reject:** The number of shackles the kill line is ahead of the last USDA button on the evisceration line.

The block will turn red if the number of shackles ahead drops to zero.



**Paw Gate:** The block will be green if the paw gate is closed and not rejecting paws. The block will be red if the gate is open. The block also contains a code showing the cause of the most recent paw rejection.



The Paw Gate is open (red) and rejecting paws for the QC Reject Button.

The S3 & QC Reject Buttons are yellow because paws are pending for rejection once they reach the paw reject gate.



## OPERATIONS AND INSTRUCTIONS

### SYSTEM CONSOLE - PAW SCREEN

#### Paw Rejection Codes:

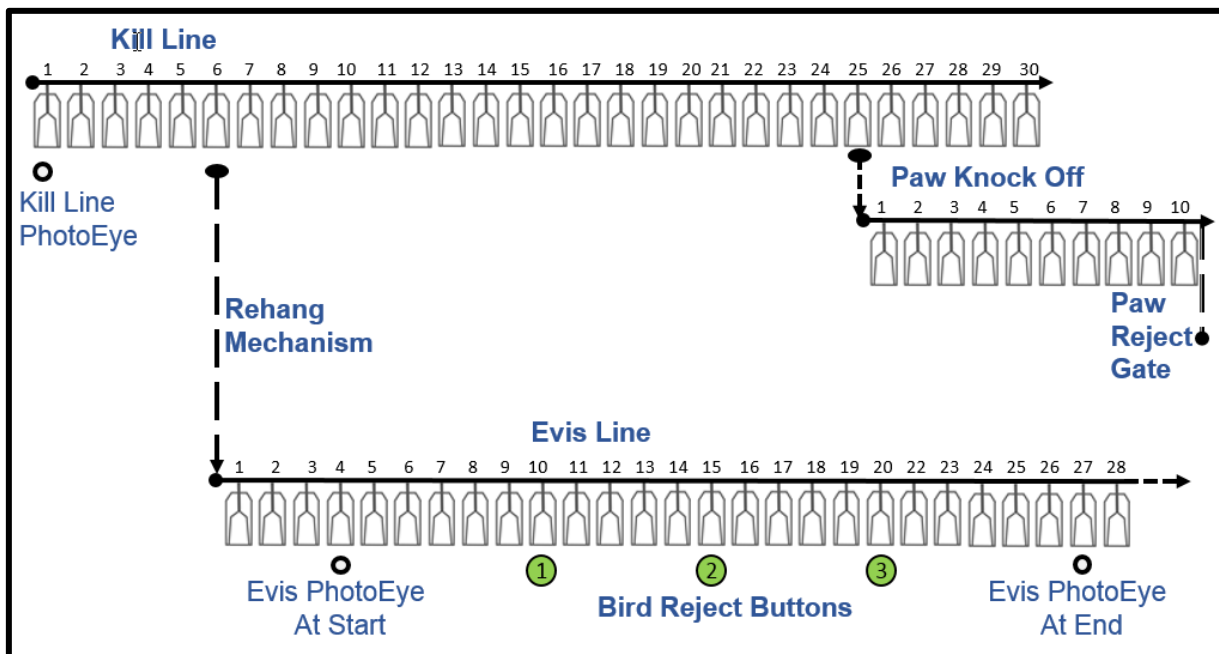
- A Number or Letter corresponding to a USDA reject button as defined for your system.
- **E'** The Evis Line was either stopped or behind the Kill Line.
- **B'** A bird was present on the Kill Line but is missing on the Evisceration Line.

#### Bird Reject Buttons Color Codes:

- **Green:** Reject Button is not pressed, and no paws are pending for rejection once they reach the paw reject gate.
- **Red:** Reject Button is currently being pressed.
- **Yellow:** Reject Button is not pressed, but paws are pending for rejection when they reach the paw reject gate.

#### Paw Tracking System Diagram

The diagram is for visual purposes only. Each plant has a unique configuration based on the number of shackles, Evis Lines, Paw Gates and USDA Bird Reject Buttons.





## System Console – Totals Report

The Totals Report shows the number of paws rejected since the beginning or the current shift or since the most recent 'Clear Totals'.

| Reject Button                 | Button Used | Before/After Count |
|-------------------------------|-------------|--------------------|
| RH                            | 1           | 11                 |
| S1                            | 0           | 0                  |
| S2                            | 2           | 16                 |
| S3                            | 1           | 11                 |
| S4                            | 1           | 11                 |
| VI                            | 2           | 22                 |
| CI                            | 0           | 0                  |
| QC                            | 2           | 22                 |
| Evis Line Down Count          | 80          | 80                 |
| Bird Missing Count            | 18          | 38                 |
| Reject All Button Count       | 0           | 0                  |
| <b>Total Paw Reject Count</b> | <b>106</b>  | <b>210</b>         |

**The Totals Report shows the number of paws rejected by each method:**

- A USDA Bird Reject button was pressed.
- The Evis Line was down or behind.
- A bird was missing on the Evis Line that was present on the Kill Line.
- The manual 'Reject All button' was pressed.

The column to the left shows the number of paw pairs rejected without including the reject Before and After counts. The column to the right shows the totals including the Before and After counts.

| Reject Button | Button Used | Before/After Count |
|---------------|-------------|--------------------|
| RH            | 1           | 11                 |

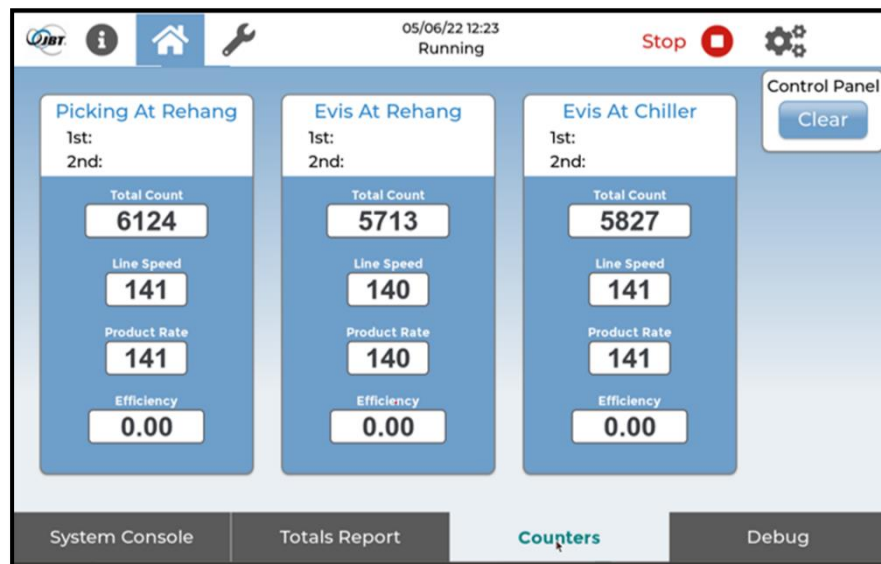
**⚙ Settings:** The number of paws to reject before and after is specified under Settings - System - Reject Settings.

### Clear Totals

Totals are automatically cleared at the beginning of each shift or can be cleared manually by touching the Clear Totals button on the screen.

## System Console – Evis Line Counter

The Evis Line Counter Screen displays information for the Primary Evisceration Line after the shift is completed.



**Total Count:** The total number of birds counted for the current shift.

**Line Speed:** The number of shackles counted for the previous minute.

**Product Rate:** The number of shackles with product (a bird) on them that were counted for the previous minute. The product rate will be displayed with a green background if the rate is at the desired rate or with a red background if below the desired rate.

**Product Efficiency:** The total bird count divided by a projected bird count for the current shift, multiplied by 100 to reach a percentage.

## OPERATIONS AND INSTRUCTIONS

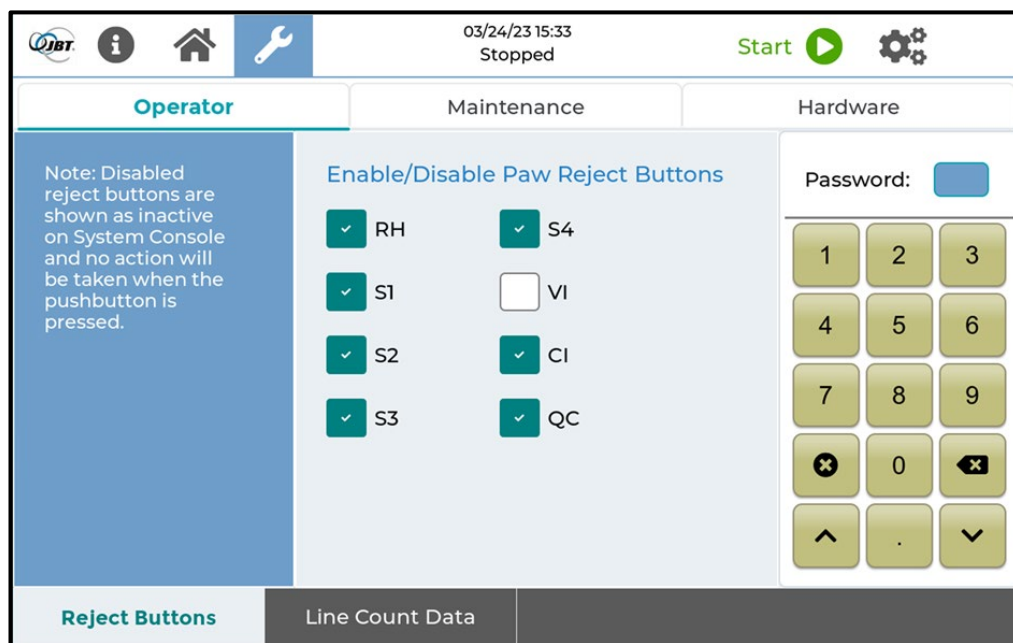
### REJECT BUTTONS

#### Tools (Wrench) – Operator – Reject Buttons



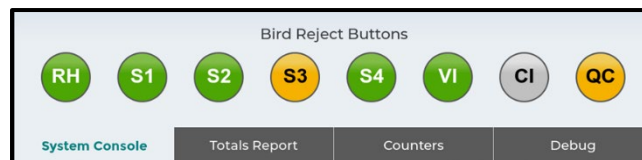
**Navigation:** Tools (Wrench) – Operator (Upper Tab) – Reject Buttons (Lower Tab)

**Security Level:** Level 1 Operator Password or above



Your screen will display the number and names of bird reject buttons that have been configured for your system.

Disabled bird reject buttons are shown as inactive on the Paw System Console. No action will be taken when an inactive pushbutton is pressed.

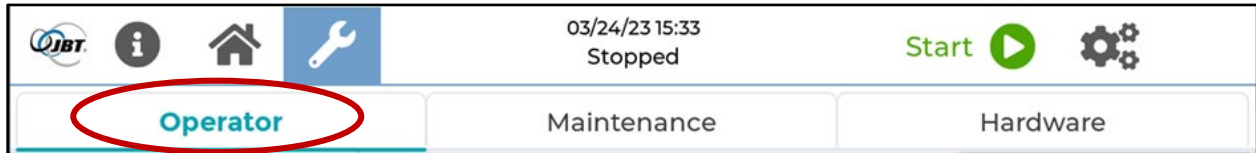


**Note:** The red 'Save' button serves as a reminder to save changes before leaving the screen.

## OPERATIONS AND INSTRUCTIONS

### LINE COUNT DATA

## Tools (Wrench) – Operator - Line Count Data



**Navigation:** Tools (Wrench) – Line Count Data (Upper Tab)

**Security Level:** Level 1 Operator Password or above

**Note:** This report is only accessible if a counter has been configured for these options.

The screenshot shows the 'Line Count Data' screen. It has a header with the JBT logo, navigation icons, date/time (03/24/23 15:24), status (Stopped), and 'Start'/'Settings' buttons. Below the navigation bar, there are three tabs: 'Operator' (selected), 'Maintenance', and 'Hardware'. The main content area is titled 'Most Recent Line Count Data' and contains a table with 8 columns: Kill Trolley Count, Kill Trolley Delta, Evis Trolley Count, Evis Shackle Count, Evis Missing Shackles, Evis Trolley Delta 2, Evis Trolley Delta 3, and Evis Trolley Delta 4. The table shows 4 rows of data, all with values of 0. Below the table, there are sections for 'Trolley Delta Current Settings' (showing 400 and 725), 'Most Recent Kill Trolley Debug Data' (showing 0), and 'Most Recent Evis Trolley Debug Data' (showing 0). At the bottom, there are 'Reject Buttons' and 'Line Count Data' tabs.

| Kill Trolley Count | Kill Trolley Delta | Evis Trolley Count | Evis Shackle Count | Evis Missing Shackles | Evis Trolley Delta 2 | Evis Trolley Delta 3 | Evis Trolley Delta 4 |
|--------------------|--------------------|--------------------|--------------------|-----------------------|----------------------|----------------------|----------------------|
| 0                  | 0                  | 0                  | 0                  | 0                     | 0                    | 0                    | 0                    |
| 0                  | 0                  | 0                  | 0                  | 0                     | 0                    | 0                    | 0                    |
| 0                  | 0                  | 0                  | 0                  | 0                     | 0                    | 0                    | 0                    |
| 0                  | 0                  | 0                  | 0                  | 0                     | 0                    | 0                    | 0                    |

**Trolley Delta Current Settings**

| Setting | Value |
|---------|-------|
| 400     | 0     |
| 725     | 0     |

**Most Recent Kill Trolley Debug Data**

| Count | Value |
|-------|-------|
| 0     | 0     |

**Most Recent Evis Trolley Debug Data**

| Count | Value |
|-------|-------|
| 0     | 0     |

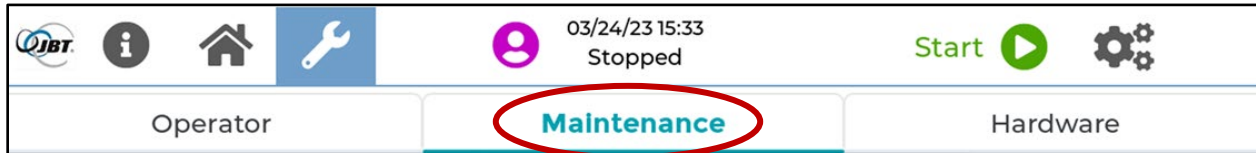
The Line Count Data screen shows the 4 most recent Trolley Counts, Shackle Counts and Trolley Deltas. See Line Count Setup for more information.

The Shackles Missing column is calculated by subtracting the Shackle Count from the Trolley Count. The Shackle Count is only used with Evisceration Lines and the Shackles Missing value indicates how many trolleys on the Evisceration Line are missing the shackle portion below that holds the bird. The shackle portion is typically made of plastic and can break off from the trolley.

## OPERATIONS AND INSTRUCTIONS

### REJECT SETTINGS

## Tools (Wrench) – Maintenance – Reject Settings



**Navigation:** Tools (Wrench) – Maintenance (Upper Tab) – Reject Settings (Lower Tab)

**Security Level:** Level 2 System Password or above

Only items configured for your system will be included on this screen.

A screenshot of the 'Reject Settings' screen in the JBT software. The interface has a top navigation bar with icons for JBT, information, home, and tools (wrench). To the right of the tools icon, it shows a user profile icon, the date and time '03/24/23 15:33', and the status 'Stopped'. Further right are 'Start' and 'Settings' buttons. Below this bar is a tabbed interface with three tabs: 'Operator', 'Maintenance' (which is highlighted), and 'Hardware'. The 'Maintenance' tab is active and contains two main sections: 'Bird Reject Settings' and 'Evis Verify Check'. The 'Bird Reject Settings' section has two columns of input fields: 'Before' and 'After'. The 'Reject Button' is set to 4 in both columns, and 'Bird Missing' is set to 5 in both columns. The 'Evis Verify Check' section has two input fields: 'Check 1' set to 0 and 'Check 2' set to 500.00. Below these fields is a 'Search for Empty Shackles' button. To the right of the 'Evis Verify Check' section is a 'Save' button with a lock icon. At the bottom of the screen is a row of five tabs: 'Reject Settings' (which is highlighted), 'Block Times', 'Shackle Counts', 'Conveyor/Gates', and an unlabeled tab. A numeric keypad is visible on the right side of the screen.

**Paw Pairs To Reject Before/After:** The number of additional paws to reject before/after for the bird a USDA inspector is currently rejecting by pressing their reject button. These values ensure that the correct paws are rejected if the USDA inspector was either a little early or a little late when they pressed their button.

**Bird Missing Paws Before/After:** The number of additional pairs of paws to reject before and after a missing bird.

**Evis Verify Check 1:** The number of shackles the Kill Line is allowed to see in a row without the Evis Line seeing any shackles go by before paws will automatically start being rejected as they pass the start of the Evis Line.

**Evis Verify Check 2:** The number of seconds the Evis Line can be stopped before all the paws on the entire Kill Line will be discarded.

#### **Search for Empty Shackles**

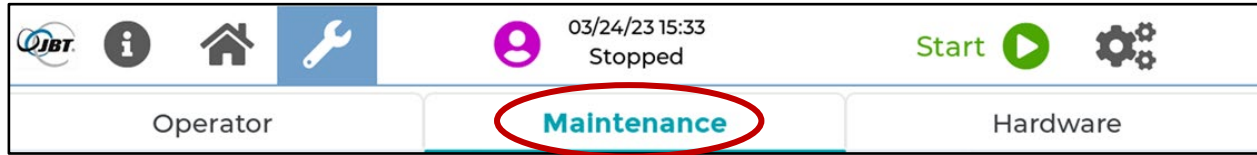
The first Search Shackles field is **Missing Rehang Search Shackles**, which is the number of shackles to search forward and backward for an empty shackle on the Kill Line when a bird is not seen at the set of Evis Photoeyes at the start of the Evis Line. If a shackle without a bird on the Kill Line is not found, then paws will be rejected since a bird must have fallen off in the Rehang mechanism.

The second Search Shackles field is **Missing Evis Search Shackles**, which is the number of shackles to search forward and backward for an empty shackle on the Evis Line when a bird is not seen at the Evis At End Photoeyes. If a shackle without a bird on the Evis Line is not found, then paws will be rejected since a bird must have fallen off the Evis line between the Evis Photoeyes At Start and the Evis Photoeyes At End.

## OPERATIONS AND INSTRUCTIONS

### BLOCK TIMES

#### Tools (Wrench) – Maintenance – Block Times



**Navigation:** Tools (Wrench) – Maintenance (Upper Tab) – Block Times (Lower Tab)  
**Security Level:** Level 2 System Password or above

A screenshot of the JBT 'Block Times' settings screen. The top bar is identical to the previous screenshot. The 'Maintenance' tab is selected, and the 'Block Times' sub-tab is active. The screen displays a table for setting minimum and maximum block times for various photoeyes. To the right of the table is a numeric keypad and a 'Save' button. At the bottom, there are four tabs: 'Reject Settings', 'Block Times' (selected), 'Shackle Counts', and 'Conveyor/Gates'.

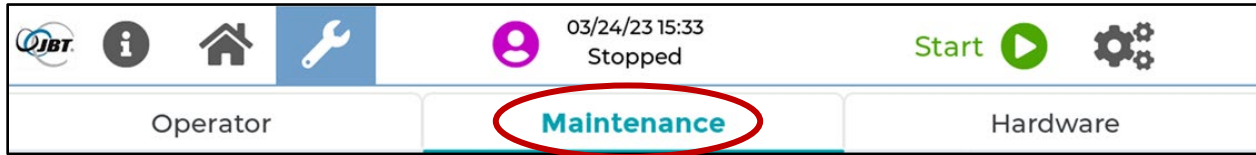
|                       | Min | Max  |
|-----------------------|-----|------|
| Kill Shaft Photoeyes: | 24  | 1000 |
| Kill Bird Photoeyes:  | 15  | 1000 |
| Evis Shaft Photoeyes: | 20  | 1000 |
| Evis Bird Photoeyes:  | 15  | 1000 |
| Bird Reject Buttons:  | 250 | 3000 |

**Block Times** control the minimum and maximum photoeye block times required for a block to be considered valid, and the minimum and maximum press times for the USDA bird reject buttons.

All the values are specified in milliseconds. Generally, the minimum and maximum block times for photoeyes that are blocked and unblocked by the shaft of a shackle are lower than the times for photoeyes blocked and unblocked by the bird hanging from the shackle.



### Tools (Wrench) – Maintenance – Shackle Counts



**Navigation:** Tools (Wrench) – Maintenance (Upper Tab) – Shackle Counts (Lower Tab)

**Security Level:** Level 2 System Password or above

The screenshot shows the 'Shackle Counts' configuration screen. It has a top navigation bar with the same icons as the previous image. Below the navigation bar, there are three tabs: 'Operator', 'Maintenance' (selected), and 'Hardware'. The 'Shackle Counts' section is active, showing various input fields and a numeric keypad. The fields are organized into two columns: 'To' and 'At Start to Reject Buttons'. The 'To' column includes 'To Evis Transfer' (24), 'To Verifies at Start' (10), 'Shackle Count From At Start' (725), and 'Shackles to Paw Knockoff' (400). The 'At Start to Reject Buttons' column includes 'RH: 20', 'S1: 300', 'S2: 400', 'S3: 500', 'S4: 600', 'VI: 700', 'CI: 810', and 'QC: 860'. A 'Save' button is located at the top right of the configuration area. At the bottom, there are four tabs: 'Reject Settings', 'Block Times', 'Shackle Counts' (selected), and 'Conveyor/Gates'.

**Shackles To Evis Transfer:** The number of shackles from the Kill Line Start Photoeye to the point where the Bird transfers from the Kill Line to the Evisceration Line at the Rehang.

**Shackles To Verifies At Start:** The number of shackles from the Bird Transfer point to the first pair of **Bird Verify** photoeyes on the Evis Line. Your installation may have only one pair of Bird Verify photoeyes or no photoeyes. The **Bird Verify** Photoeyes provide for discarding the paws if a Bird was present when it passed the Kill Line Photoeyes but is missing when it passes the Bird Verify photoeyes. Typically, the Bird Verify Photoeyes will be near the beginning of the Evisceration Line to discard paws when a bird falls off at the Rehang.

**Shackles To Verifies at End:** The number of shackles from the Verifies at Start Photoeye to the Verifies At End Photoeye, after the last Bird Reject Button.



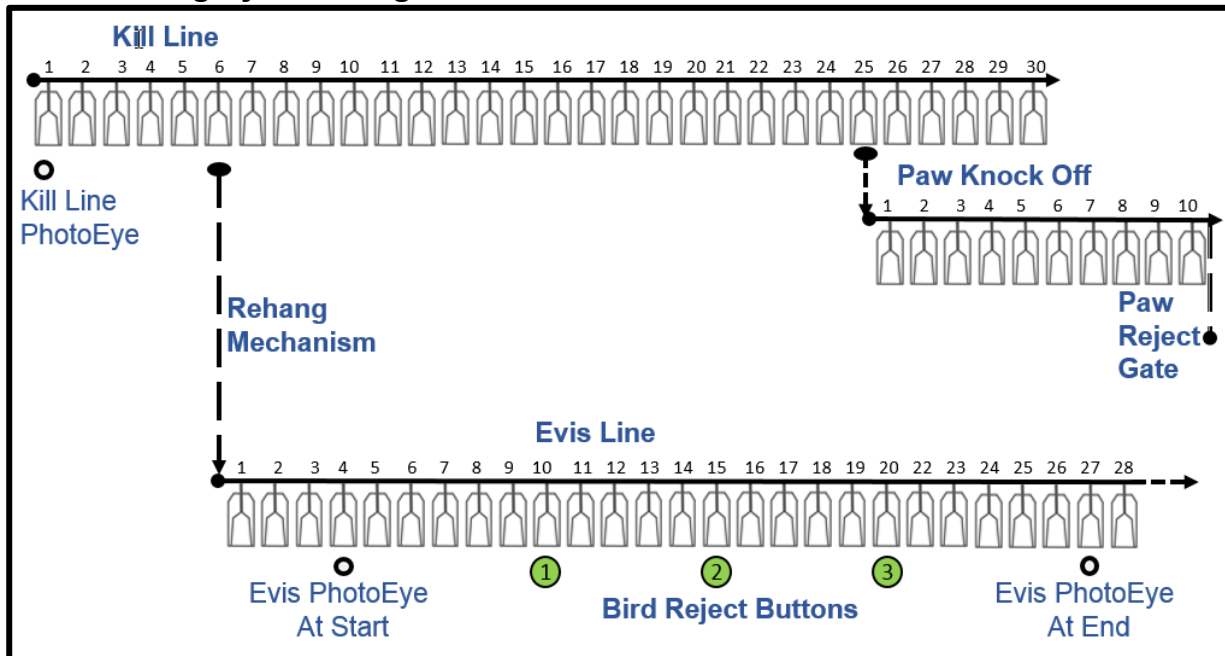
## OPERATIONS AND INSTRUCTIONS

### SHACKLE COUNTS

**Shackles To Paw Knockoff:** The number of shackles from the Kill Shaft Photoeye to the end of the Kill Line where the paws either fall on the Paw Conveyor or exit the system if there is no paw conveyor.

**Shackles To Evis Bird Reject Buttons:** The number of shackles from the Evis Photoeye at Start (Bird Verify Photoeyes) to the Bird Reject buttons.

#### Paw Tracking System Diagram

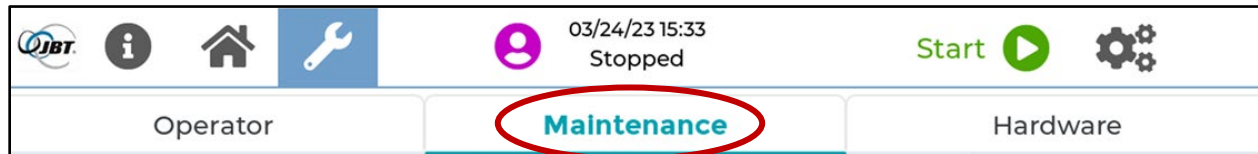


The diagram above is for visual purposes only. Each plant has a unique configuration based on the number of shackles, Evis Lines, Paw Gates and USDA Bird Reject Buttons.

## OPERATIONS AND INSTRUCTIONS

### CONVEYORS/GATES

## Tools (Wrench) – Maintenance – Conveyor/Gates



**Navigation:** Tools (Wrench) – Maintenance (Upper Tab) – Conveyor/Gates (Lower Tab)

**Security Level:** Level 2 System Password or above

A screenshot of the JBT maintenance interface showing the 'Conveyor/Gates' settings. The top bar is identical to the previous screenshot. The 'Maintenance' tab is selected, and the 'Conveyor/Gates' sub-tab is active. The interface is divided into three main sections: 'Conveyors', 'Gate Open Times', and a numeric keypad. The 'Conveyors' section includes '1st Conveyor Travel Time' (4.00), 'Paw Conveyor Settings' (Travel Time 1 (Fast): 360.00, Travel Time 2 (Med): 480.00, Travel Time 3 (Slow): 600.00), and 'Shackles to Slow Speed' (Shackles to Slow Speed 1: 50, Shackles to Slow Speed 2: 25). The 'Gate Open Times' section includes 'Paw Reject Gate' (1.00) and 'Bird Missing Gate' (1.00). The numeric keypad has buttons for digits 1-9, 0, a backspace key, and up/down arrow keys. A 'Save' button is located at the top right of the keypad area. At the bottom, there are four tabs: 'Reject Settings', 'Block Times', 'Shackle Counts', and 'Conveyor/Gates' (which is highlighted).

**First Conveyor Travel Time:** The number of seconds it takes a pair of paws to travel across the first fixed speed conveyor. Your installation may not have a fixed speed conveyor.

**Paw Conveyor Travel Time 1:** The number of seconds, at normal conveyor speed, required for a pair of paws to travel to the Paw Reject Gate on the variable speed Paw Conveyor.

**Paw Conveyor Travel Time 2:** The number of seconds at the first slow speed, required for a pair of paws to travel to the Paw Reject Gate.

**Paw Conveyor Travel Time 3:** The number of seconds at the slowest conveyor speed, required for a pair of paws to travel to the Paw Reject Gate.

## OPERATIONS AND INSTRUCTIONS

### CONVEYORS/GATES

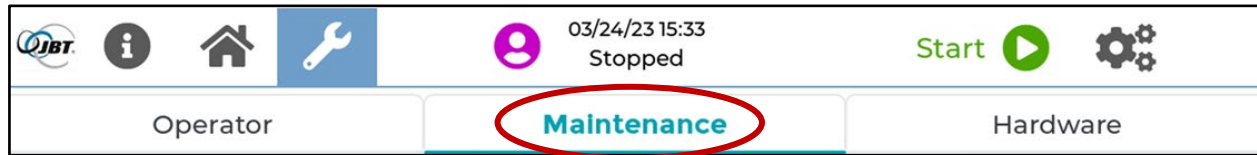
**Shackles To Slow Paw Conveyor:** The lowest number of shackles the Kill Line and Paw Conveyor can be ahead of the last USDA inspection station before the paw conveyor is slowed. Slowing the paw conveyor gives more travel time for the paws on the paw conveyor hopefully allowing the Evis Line to catch back up to the Kill Line. The system supports two thresholds to slow the paw conveyor.

**Reject Gates Open Time:** The number of seconds the Bird Missing and Paw reject gates should stay open when rejecting paws.

## OPERATIONS AND INSTRUCTIONS

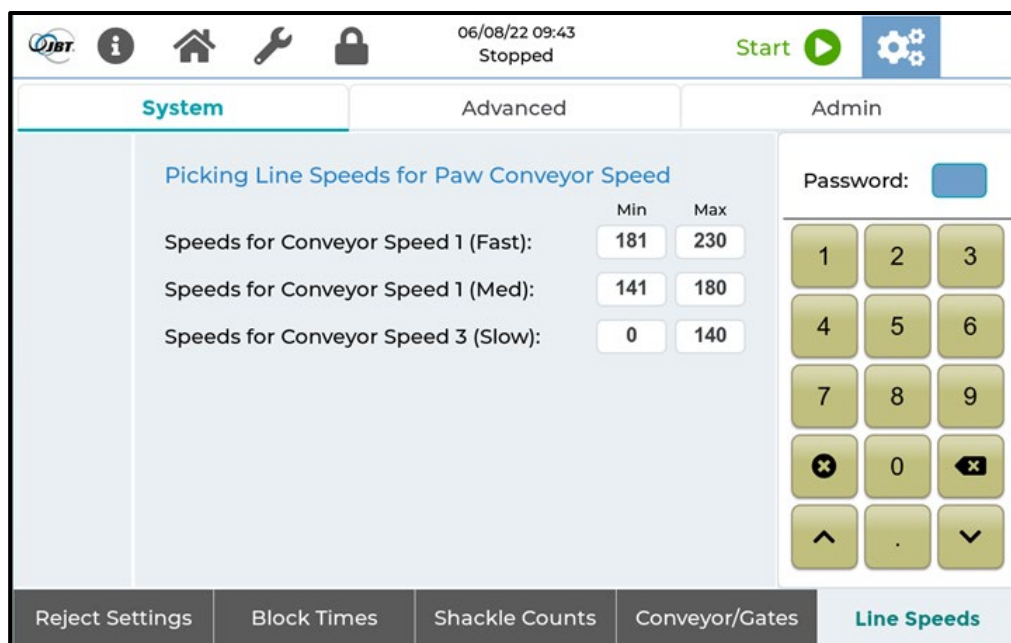
### LINE SPEEDS

## Tools (Wrench) – Maintenance – Line Speeds



**Navigation:** Tools (Wrench) – Maintenance (Upper Tab) – Line Speeds (Lower Tab)

**Security Level:** Level 2 System Password or above

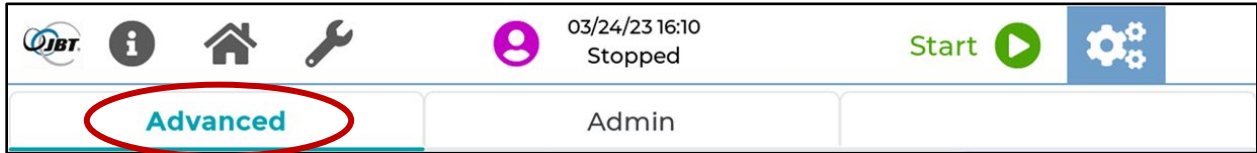


**The Line Speed tab is only visible when 'Use Kill Line Speeds for Paw Conveyor Speed' is enabled.** See Settings – Advanced – Controller Setup for more information.

If using a variable speed conveyor, the conveyor can be slowed either by shackle-count (see Settings (Cogs icon) - System (upper tab) – Conveyors/Gates (lower tab) or based on the kill line speed.

The Line Speeds page supports controlling the paw conveyor speed based on the range of min / max speeds.

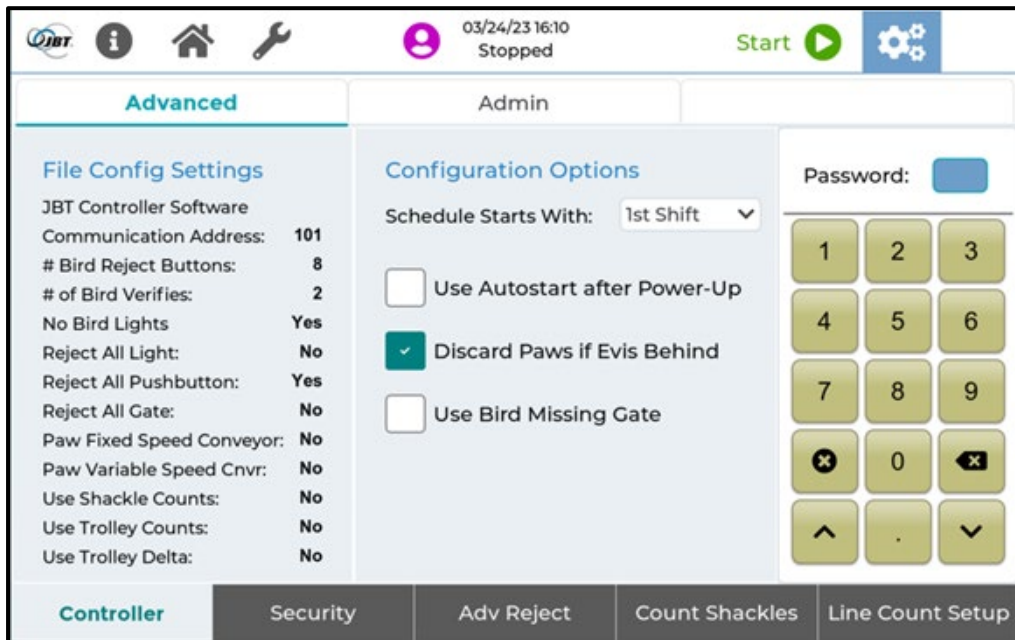
### Adv Settings (Cogs) – Advanced - Controller



**Navigation:** Adv Settings (Cogs) – Advanced (Upper Tab) – Controller (Lower Tab)

**Security Level:** Level 3 Advanced Password or above

These values are usually set during the initial installation.



**File Config Settings:** Configuration settings that are specified in your plant's configuration file.

**Communication Address:** Assigned number from 101 to 110 to uniquely identify this JBT Controller when communicating with the computer running the **JBT Efficiency Management System** program in a nearby office.

### Configuration Options

**Schedule Starts with 3<sup>rd</sup> Shift:** Select the appropriate option for the plant's schedule.

**Use Autostart after Power-up:** Select to have the software automatically start after a power-outage or reset.

## OPERATIONS AND INSTRUCTIONS

### ADVANCED SETTINGS - CONTROLLER

**Discard Paws when Evis Behind:** Enables automatically discarding paws when the Evis Line falls behind the Kill Line. Paws which are linked to Birds on the Evis Line will reach the end of the Paw Conveyor before the associated bird has been inspected. If this occurs, any paws that reach the end of the Paw Conveyor before the bird has passed the last reject button will be automatically rejected.

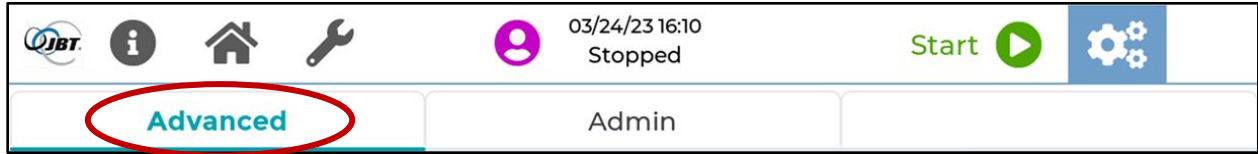
**Use Bird Missing Gate:** Enables the Bird Missing Gate functionality. The Missing Gate will be displayed on the System Console as shown below.

**Use Kill Line Speed to Set Paw Conveyor Speed:** This option is only available for variable speed conveyors. The conveyor can be slowed by shackle-count as shackles slow down, or when the kill line speed is in a certain range. Enabling this option allows a range of min/max speeds for each conveyor speed. (See Settings – System – Line Speeds)

Note: The Line Speed Tab is only visible when this option is enabled.

**Note:** This option will only be available if your configuration supports enabling this feature.

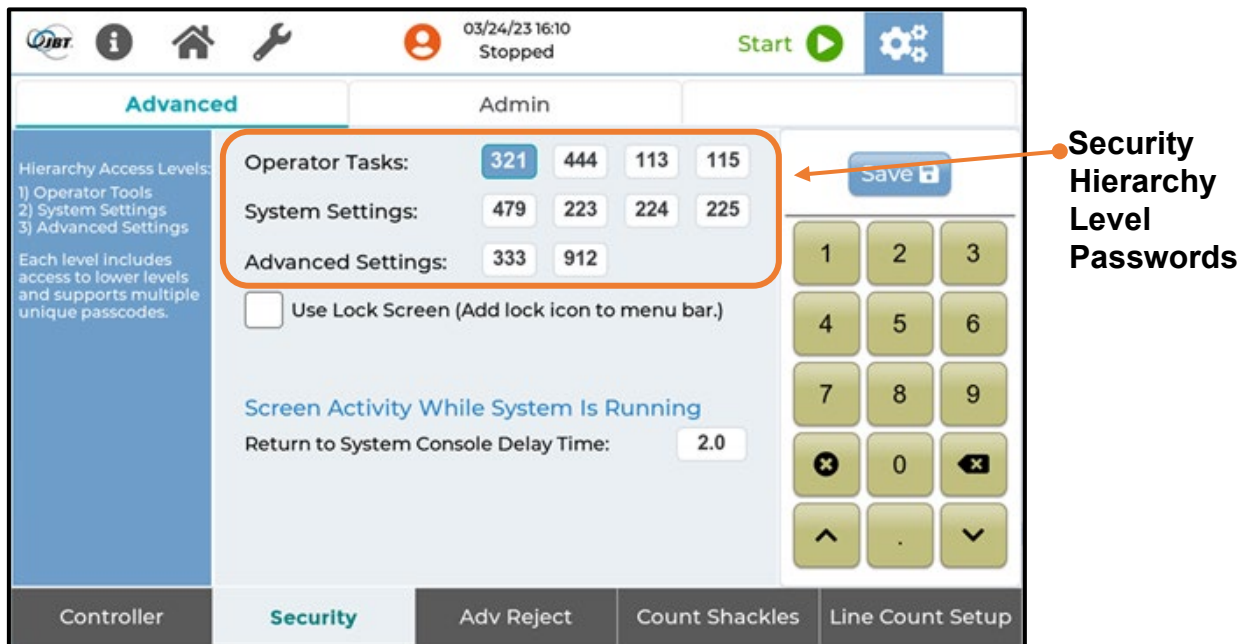
### Adv Settings (Cogs) – Advanced - Security



**Navigation:** Adv Settings (Cogs) – Advanced (Upper Tab) – Security (Lower Tab)

**Security Level:** Level 3 Advanced Password

The Gen 2 Security has changed from Gen 1's individual passwords managed externally, to a **hierarchy style managed by the plant.**



### Security Hierarchy

Passwords are evaluated by security level, and higher level passwords work on lower level tasks. For example, an Operator password is only valid on Operator tasks, were as System level password works on both Operator and System levels, and Advanced passwords work on Operator, System and Advanced tasks.

**Level 1 Operator** passwords are required for Operator type activity such as enabling reject buttons

**Level 2 System Setting** passwords are required for system configuration changes. These passwords should only be given to those who have the knowledge and authority to change system settings. Level 2 password also work for Level 1 Tasks.

## OPERATIONS AND INSTRUCTIONS

### ADVANCED SETTINGS - SECURITY

**Level 3 Advanced Setting** passwords are required for advanced system settings that should only be done by trained individuals. Level 3 passwords also work for Level 1 & Level 2 Tasks.

#### Use Lock Screen

The Lock Screen setting controls the appearance of a lock icon on the menu bar, shown above. Touching the lock icon disables the menu bar and shows a 'System is Locked' message screen. Unchecking this setting removes the lock icon from the header menu.

#### Unlock Screen Passcode

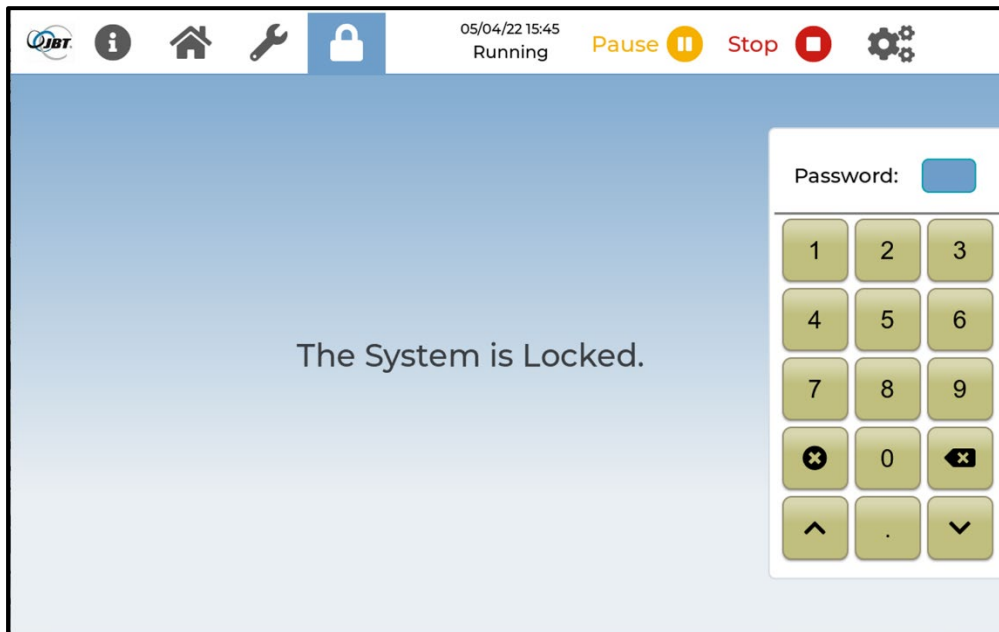
Enter a passcode to unlock the system. Unlock using the unlock passcode **or any other valid passcode**. The unlock passcode, and all configured passcodes can be used to unlock the system.

#### Return to System Console Delay Time

If the system is running, the home screen/system console is not the active screen, and there has been no user activity, the system will return to the home screen after the configured amount of time has passed.

#### System Console Screen Lock Delay Time

If the system console is being displayed, and there has been no user activity, the screen locks after the configured minutes





## Plant Managed Passwords

Gen 2 provides a designated login to manage passwords at the plant level. The Security Admin password only works on the Security Screen and can't be used to perform system tasks. Operator and System levels support up to 4 different passwords, and Advanced Level support 2 passwords.



## Invalid Passwords

Passwords must be three digits and can not start with 0. An invalid password entry will be changed back to the original value and a red border added to the textarea block to alert the user that the entered password was rejected.

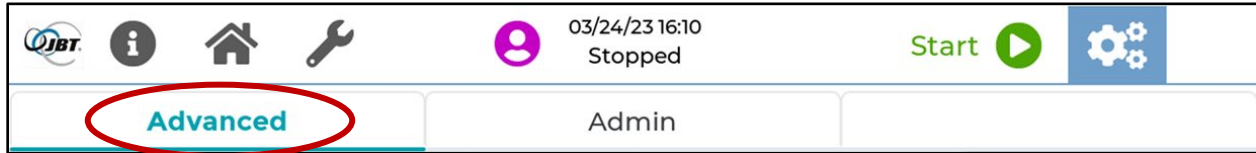
## When to Assign 4 Different Passwords

The Operator Tasks example above shows four different passwords which can be given to the various operators as needed.

## When to Assign Same Passwords

For plants transitioning from Gen1 to Gen2, it's possible that only one or two people need a system level password, or everyone is used to using a certain password. **Using the same password allows the plant to continue with their Gen1 password and assign other passwords at their convenience.**

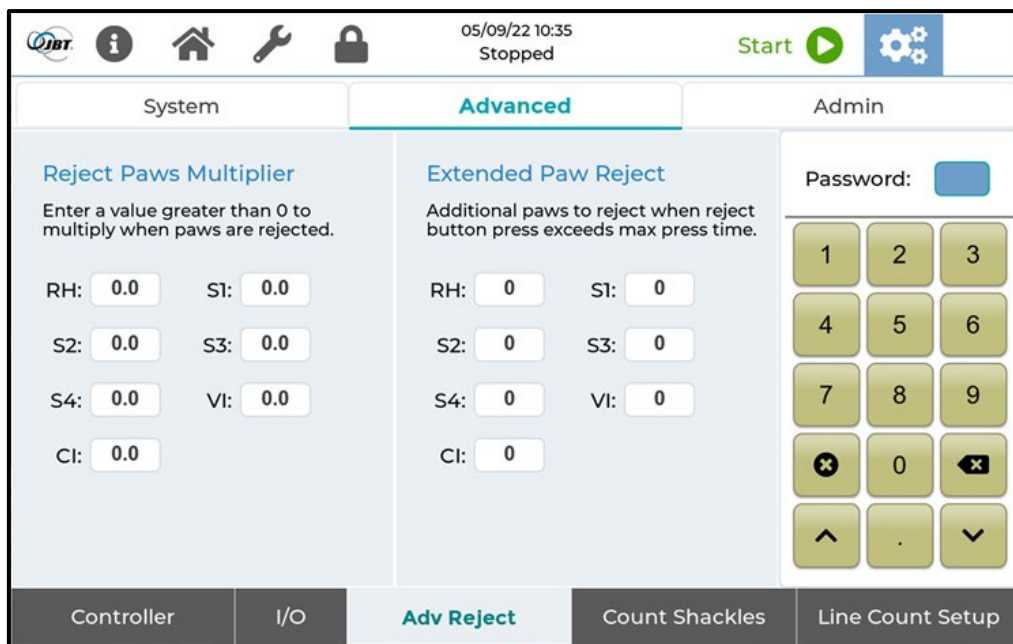
## Adv Settings (Cogs) – Advanced – Adv Reject Settings



**Navigation:** Adv Settings (Cogs) – Advanced (Upper Tab) – Adv Reject (Lower Tab)

**Security Level:** Level 3 Advanced Password or above

Due to the high impact of these settings, changes should not be made to these settings without guidance from a JBT Technician.



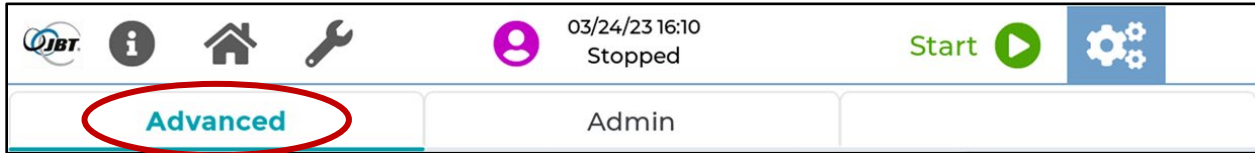
**Reject Paws Multiplier:** The multiplier is used to calculate the number of paws to reject before and after the reject button is pressed. If the before/after count is set to 5/5 (see System Setup 3), a multiplier of 2 would reject 10/10 paws before/after each time the associated reject button is pressed.

**Extended Reject Paws After:** The extended reject option only takes effect when the associated button is held longer than the max press time (see Settings - System -Reject Settings).

If the operator holds the button longer than the max press time, the following occurs:

- 1) Paws flagged for discard during normal button press will be canceled.
- 2) The specified number of paws, starting when the button was originally pressed, will be flagged for discard.

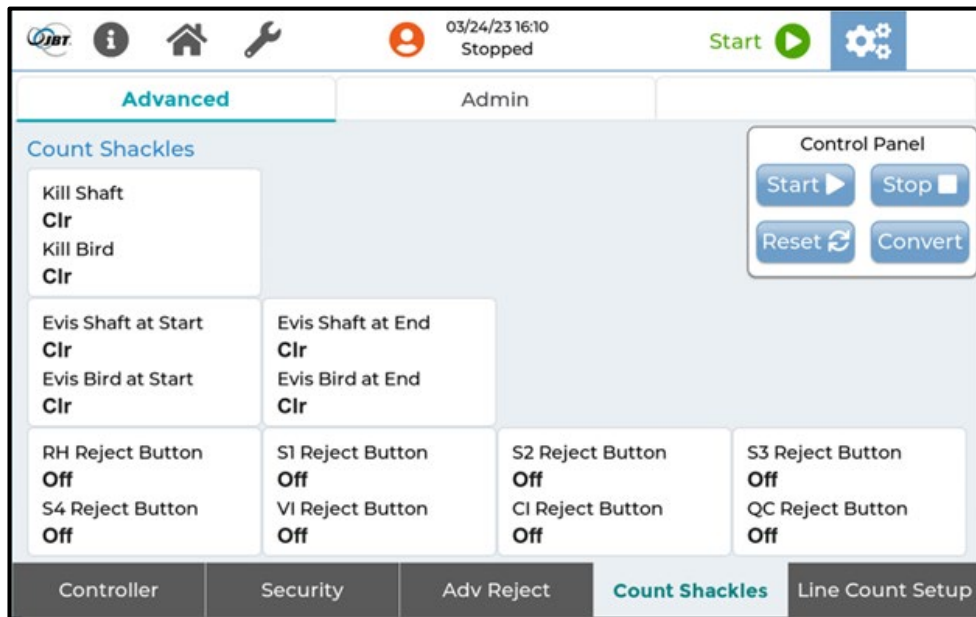
## Adv Settings (Cogs) – Advanced - Count Shackles



**Navigation:** Adv Settings (Cogs) – Advanced (Upper Tab) – Count Shackles (Lower Tab)

**Security Level:** Level 3 Advanced Password or above

The **Count Shackles Test** counts the number of shackles between the photoeyes and bird reject buttons that are installed along the kill line and evisceration line. This test is normally only used during the initial installation and configuration of the **MARS Paw Tracking System** at your plant, or when shackle counts between photoeyes and buttons have changed.



**Count Shackles** requires hanging and tracking a single bird as it passes in front of the various photoeyes and reject buttons. This can only be executed during non-production periods.

### Requirements Before Running Initial Test:

- 1) Manually count number of shackles from kill shaft photoeye to evis transfer point and enter the values on **Shackles To Evis Transfer** on **System – Shackle Count**.
- 2) Configure Min / Max times for the photoeyes and buttons on **System - Blocktimes**.
- 3) **Save** changes.

## OPERATIONS AND INSTRUCTIONS

### ADVANCED SETTINGS - COUNT SHACKLES

#### Start the Test

Touch the Start button and hang a single bird just upstream of the Kill Shaft and Kill Bird Photoeyes.

#### Testing Sequence

1) **Bird is seen by Kill Bird Photoeye.**

Kill shackles will start counting up as they pass the kill shaft photoeye.

2) **Bird transfers to Evis Line and is seen by Evis At Start Photoeye.**

Number of kill shackles will be displayed under Evis Bird Photoeye1 and will start counting evis shackles as they pass the Evis Shaft Photoeye1.

3) **Bird travels down Evis line and is seen by Evis At End Photoeye.**

When the bird is seen at each of these photoeyes, the number of evis shackles and kill shackles that have been counted will be displayed under each bird photoeye.

The number of evis shackles is displayed on the left and the number of kill shackles is displayed to the right.

4) **Bird passes in front of each reject button.**

**Note:** Someone must touch the reject button so the system can track the number of evis shackles and kill shackles which have been counted for that button.

5) **After Bird has passed all photoeyes and buttons,**

Touch **Stop** to prevent the shackles from continuing to count.

The screen at the top of the next page shows a sample completed test.

| Count Shackles                     |                                        |                                        |
|------------------------------------|----------------------------------------|----------------------------------------|
| Kill Shaft Photoeye<br>Blocked 437 | Evis Shaft Photoeye1<br>Clear 402 437  | Evis Shaft Photoeye2                   |
| Kill Bird Photoeye<br>Clear 0      | Evis Bird Photoeye1<br>Clear 0 35      | Evis Bird Photoeye2                    |
| Evis Start Photoeye<br>Blocked     | Adjust Shaft Photoeye1<br>Clear        | Adjust Shaft Photoeye2<br>Clear        |
|                                    | Adjust Bird Photoeye1<br>Clear 270 305 | Adjust Bird Photoeye2<br>Clear 384 419 |
| Reject Button 1<br>Off 292 327     | Reject Button 2<br>Off 315 350         | Reject Button 3<br>Off 335 370         |
| Reject Button 4<br>Off 367 402     | Reject Button 5                        | Reject Button 6                        |
| Back                               | Start                                  | Stop                                   |
| Reset                              | Convert                                |                                        |

The values displayed (left) do not include the number of evis shackles from the evis transfer point to the first set of photoeyes on the evis line.

Touch **Convert** to update the evis shackle counts to include the additional shackles. The final converted values will be shown in white with a black background as shown below.

## OPERATIONS AND INSTRUCTIONS

### ADVANCED SETTINGS - COUNT SHACKLES

Copy the converted values from the test to System Setup 2.

| Count Shackles                                                                         |                                                              |                                                             |                                                                              |
|----------------------------------------------------------------------------------------|--------------------------------------------------------------|-------------------------------------------------------------|------------------------------------------------------------------------------|
| <b>Kill Shaft Photoeye</b><br><input type="radio"/> Blk                                | <b>Evis Shaft At Start</b><br>Clr 402 437                    |                                                             |                                                                              |
| <b>Kill Bird Photoeye</b><br>Clr                                                       | <b>Evis Bird At Start</b><br>Clr 20 35                       | Shackles To Verifies At Start                               |                                                                              |
| <b>Evis Start Photoeye</b><br>Clr                                                      | <b>Evis Shaft At End</b><br><input type="radio"/> Blk        |                                                             |                                                                              |
|                                                                                        | <b>Evis Bird At End</b><br><input type="radio"/> Blk 290 305 | Shackles To Verifies At End                                 |                                                                              |
| <b>Reject Button R</b><br><input checked="" type="radio"/> On 312 327<br>Reject Button | <b>Reject Button 1</b><br><input type="radio"/> Off 335 350  | <b>Reject Button 2</b><br><input type="radio"/> Off 355 370 | <b>Reject Button 3</b><br><input type="radio"/> Off 375 400<br>Reject Button |
| Shackles To Reject Buttons                                                             |                                                              |                                                             |                                                                              |

The test should be repeated several times to verify consistent results. To run the test again touch **Reset** to clear values and then **Start**.

### Shackles To Paw Knockoff

✂ The Shackles To Paw Knockoff value can be difficult to count manually as the number of shackles often exceeds 500 shackles.

## OPERATIONS AND INSTRUCTIONS

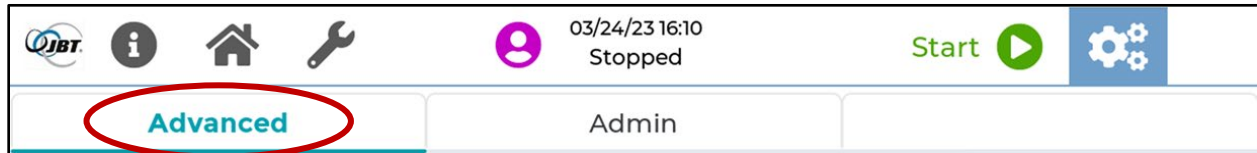
### ADVANCED SETTINGS - COUNT SHACKLES

**Counting Help:** Use one of the reject buttons to count by pressing the reject button when the paws reach the paw knockoff. Normally the reject button is pressed when the bird passes, however the above method is useful for counting purposes because pressing the reject button will record the number of evis and kill shackles that have been counted under the button.

The number of kill shackles is the number on the right under the button and is the value that should be entered for **Shackles To Paw Knockoff** on System Setup 3.

After running the test a sufficient number of times to verify consistent results, and entering the resulting values System Setup 2-3, you should be ready to start verifying the correct operation of the **MARS Paw Tracking System**.

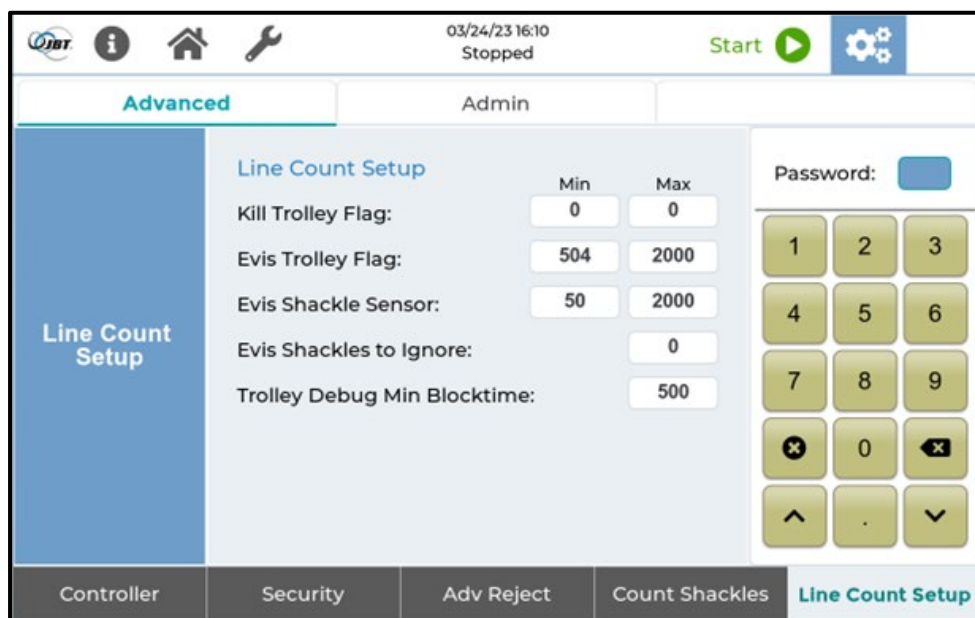
### Adv Settings (Cogs) – Advanced – Line Count Setup



**Navigation:** Adv Settings (Cogs) – Advanced (Upper Tab) – Line Count Setup (Lower Tab)

**Security Level:** Level 3 Advanced Password or above

Several options are available for testing the system at various speeds. These tests are normally only used during the initial installation and configuration of the **JBT Paw Tracking System** at your plant.



**Trolley Flag:** Minimum and maximum time that the shaft photoeye must be blocked by the special trolley that signals that a complete revolution of the chain has occurred. This special trolley has a piece of metal welded to it so that it blocks the shaft photoeye longer than the other trolleys. The Trolley Debug Min Block Time value below along with the Trolley Debug Data (Advanced Setup Menu) can be used to determine how long the special trolley blocks the shaft photoeye.

**Shackle Sensor:** Minimum and maximum time that the special shackle photoeye on the Evisceration Line must be blocked to count a valid trolley with a shackle intact below it. The Hardware Test screen can be used while the Evisceration Line is running to observe typical shackle block times.



## OPERATIONS AND INSTRUCTIONS

### ADVANCED SETTINGS - LINE COUNT SETUP

**Shackles to Ignore:** This value is usually set to 0 to specify that every valid block of the shackle photoeye discussed above will count toward the Shackle Count. Some plants may have chains that cause the shackle photoeye to block a second time when every shackle goes by. In this case this value would be set to 1 specifying that only every other valid block of the shackle photoeye should count toward the Shackle Count.

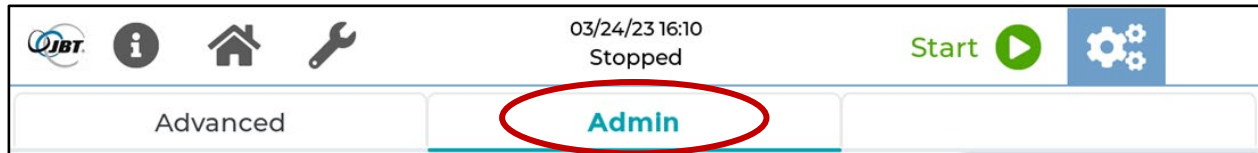
**Trolley Debug Min Blocktime:** This value should be set to slightly less than how long the special Trolley Count trolley will block the shaft photoeye. This value should also be set higher than how long a typical trolley would block. The special trolley for the Trolley Count that indicates a whole revolution of the chain has occurred will block longer than a typical trolley because of the extra metal that has been welded to it. The actual block time will be stored in memory for the last 6 trolleys that have gone by where the block time exceeded this setting. The 6 most recent trolley block times that exceeded this setting can then be accessed on the Trolley Debug Data screen from the Advanced Setup menu. The 6 values shown can then be used to set the Trolley Debug Min Block time correctly.



## OPERATIONS AND INSTRUCTIONS

### ADMIN SCREENS - GENERAL

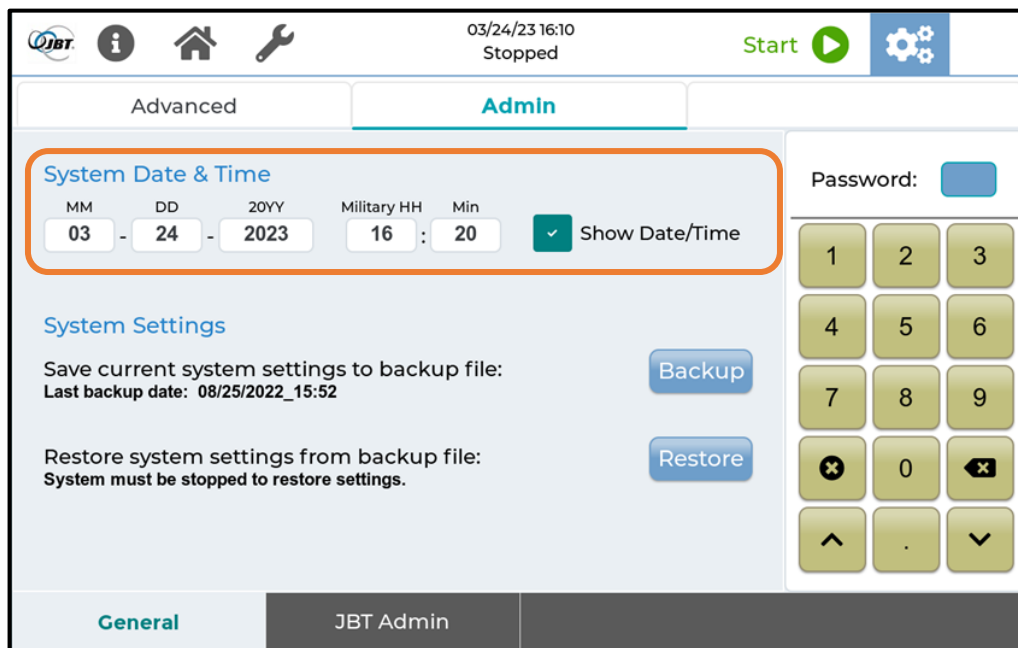
#### Adv Settings - Admin - General



**Navigation:** Adv Settings (Cogs) – Admin (Upper Tab) – General (Lower Tab)

**Security Level:** Level 3 Advanced Password or above

The General Admin screen contains settings and tools that are unrelated to your system's configuration or operation.



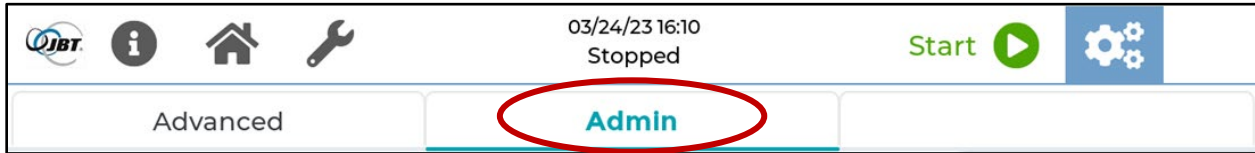
**System Date & Time:** The JBT controller has a battery-backed up clock for maintaining the correct Date and Time when the controller is not connected to the internet. Enter the correct Date and Time after logging in with your password. Then press the Save button.

**Exit to Operating System:** Stop the system and exit the JBT software to the Operating System.

## OPERATIONS AND INSTRUCTIONS

### ADMIN SCREENS - JBT ADVANCED ADMIN

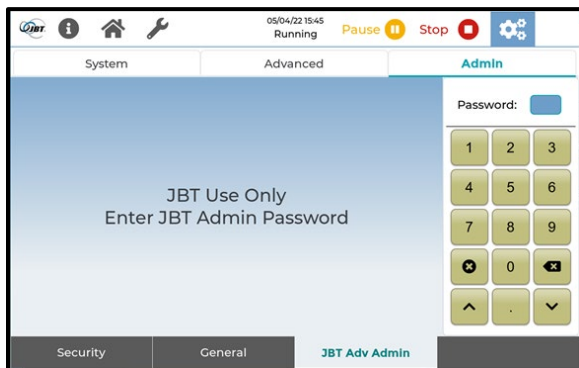
#### Adv Settings - Admin – JBT Adv Admin



**Navigation:** Adv Settings (Cogs) – Admin (Upper Tab) – JBT Adv Admin (Lower Tab)

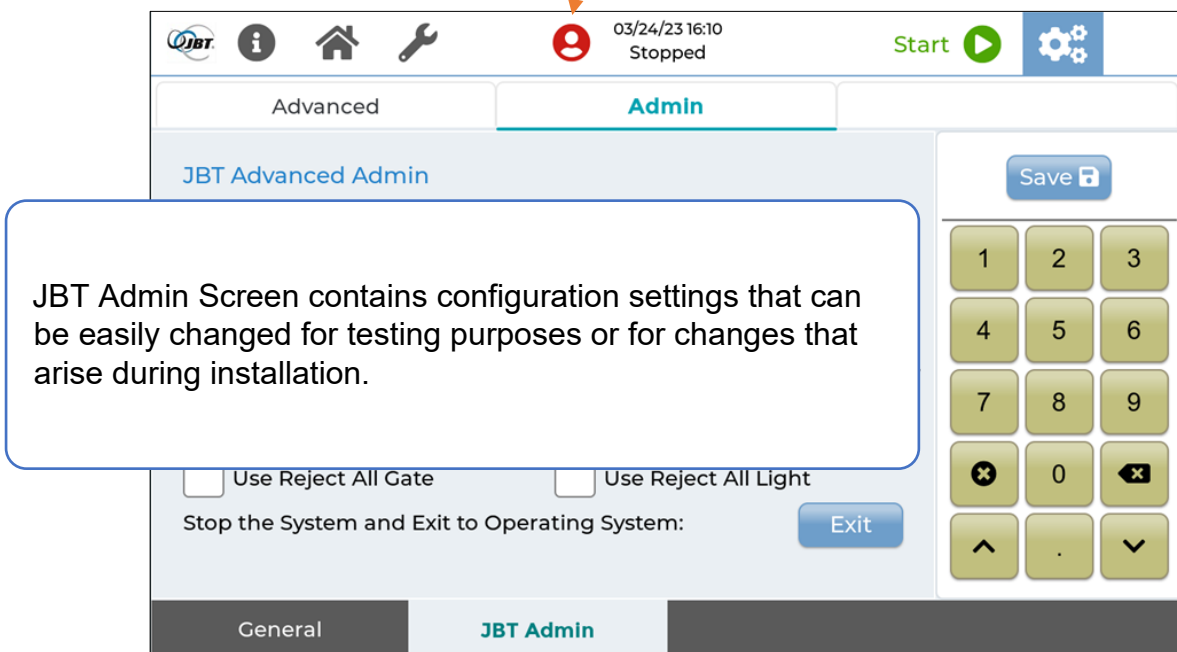
**Security Level:** JBT Super User (JBT Use Only)

The JBT Admin screen contains settings and tools that are used exclusively by a JBT Technician during installation.



JBT Admin is logged in and passwords are disabled.

Touch icon to logout as JBT Admin.



JBT Admin Screen contains configuration settings that can be easily changed for testing purposes or for changes that arise during installation.

## SANITATION

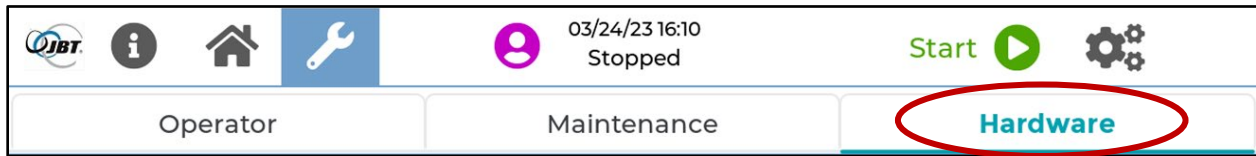
### SANITATION PROCEDURES

There are no sanitation procedures for this manual.

## MAINTENANCE AND TROUBLESHOOTING

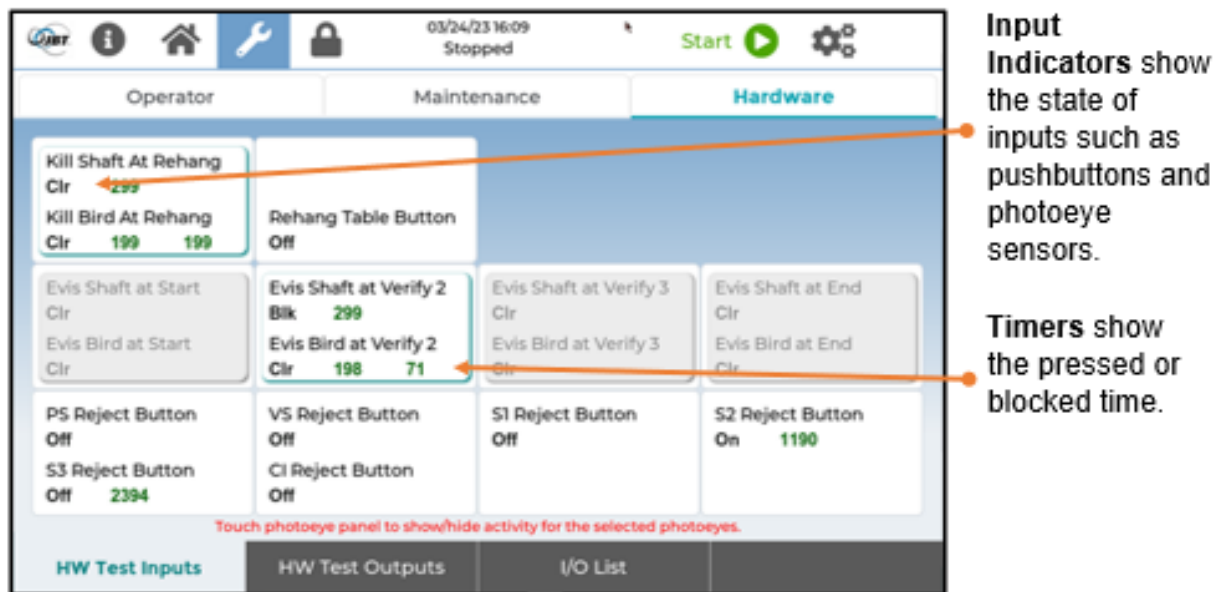
### HARDWARE TEST INPUTS

## Tools (Wrench) – Hardware - HW Test Inputs



**Navigation:** Tools (Wrench) – Hardware (Upper Tab) – HW Test Inputs (Lower Tab)  
**Security Level:** Level 2 System passcode or above

Your Hardware Test screen will reflect the inputs and outputs for your system.



### Verify Inputs

The input buttons are shown as small circles which reflect the current state of the input.

Pushbutton States:  

Photoeye Sensors:  

### Product Photoeye Block Times

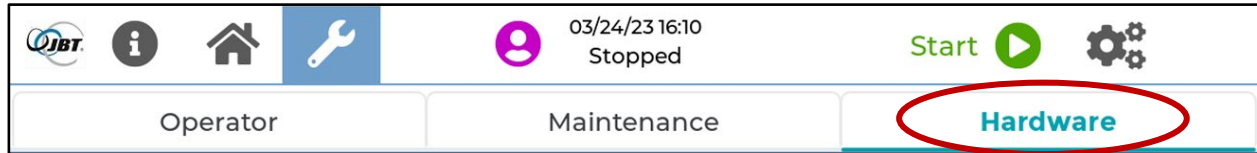
- The value on the left is the total amount of time the photoeye was blocked by product (a bird)
- The value on the right is the amount of time the product photoeye was blocked **while the corresponding shaft photoeye was also blocked**.

 **Settings:** The min and max block times are configured in System - Blocktimes

## MAINTENANCE AND TROUBLESHOOTING

### HARDWARE TEST INPUTS

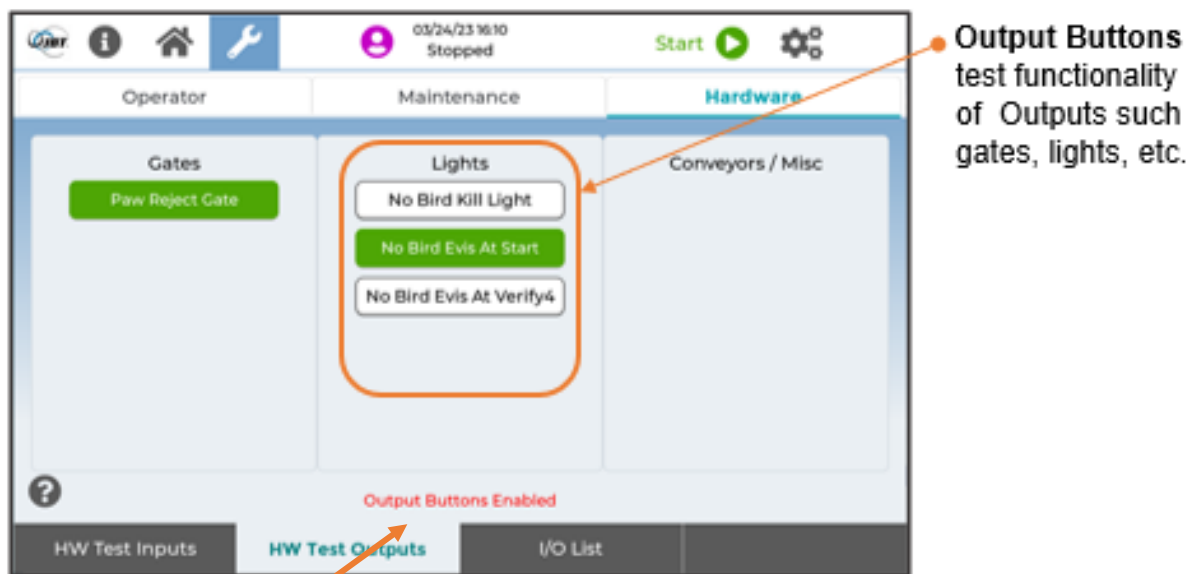
#### Tools (Wrench) – Hardware - HW Test Inputs



**Navigation:** Tools (Wrench) – Hardware (Upper Tab) – HW Test Inputs (Lower Tab)

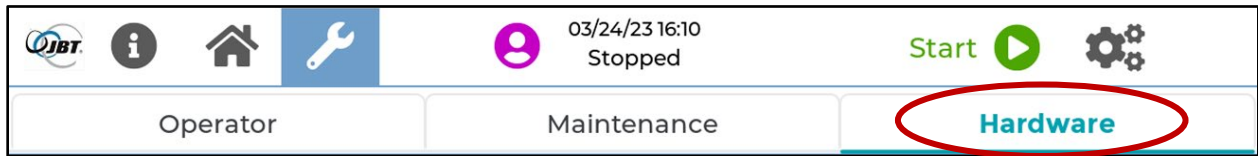
Use the output buttons to test the signal used to control hardware such as gates and external lights. The output buttons also reflect the state of the output while the system is running. Touch the appropriate button to verify the correct response.

See Settings (Cogs) – Advanced (Upper Tab)- I/O List (Lower Tab) for I/O assignments.



| Status Message                           | Desc/Required Action                                                                                    |
|------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Output Buttons Disabled (System Running) | Testing outputs is not allowed while the system is running. Stop the system to test outputs.            |
| Login to Enable Output Buttons           | The system is stopped but a password is required before testing the output buttons. Touch Login button. |
| Output Buttons Enabled                   | Login was successful. Ready to test Output buttons.                                                     |

### Tools (Wrench) – Hardware - I/O List



**Navigation:** Tools (Wrench) – Hardware (Upper Tab) – I/O List (Lower Tab)

The I/O (Inputs/Outputs) screen provides a convenient list of the inputs and outputs configured for your plant's installation. Testing these inputs/outputs is performed on Hardware Test.

| Operator              |    | Maintenance        |    | Hardware               |    |
|-----------------------|----|--------------------|----|------------------------|----|
| <b>Inputs</b>         |    | <b>Inputs</b>      |    | <b>Outputs</b>         |    |
| Kill Shaft At Rehang: | 1  | Reject Button S4:  | 11 | Paw Gate:              | 21 |
| Kill Bird At Rehang:  | 2  | Reject Button VI:  | 12 | Bird Missing Gate:     | 22 |
| Evis At Start Shaft:  | 3  | Reject Button CI:  | 13 | No Bird Kill Light:    | 23 |
| Evis At Start Bird:   | 4  | Reject Button QC:  | 14 | No Bird Evis Verify 1: | 24 |
| Evis Verify 4 Shaft:  | 5  | Reject All Button: | 15 | No Bird Evis Verify 4: | 25 |
| Evis Verify 4 Bird:   | 6  |                    |    |                        |    |
| Reject Button RH:     | 7  |                    |    |                        |    |
| Reject Button S1:     | 8  |                    |    |                        |    |
| Reject Button S2:     | 9  |                    |    |                        |    |
| Reject Button S3:     | 10 |                    |    |                        |    |
| HW Test Inputs        |    | HW Test Outputs    |    | I/O List               |    |

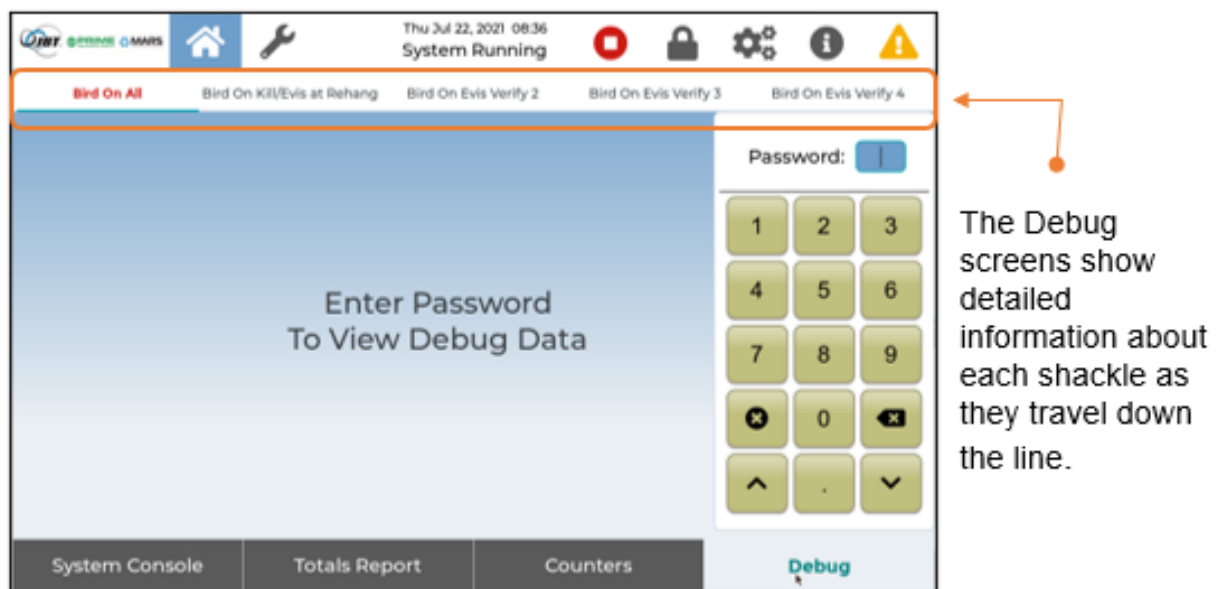
Your I/O list will be customized for your plant's configuration.

## System Console – Debug Tools



**Navigation:** System Console (Home) – Debug (Lower Tab)

**Security Level:** Level 1 Expert Password



The Debug screens show detailed information about each shackle as they travel down the line.

Shackle Data is available for the following locations on the line.

- Bird On All (Locations)
- Bird On Kill/Evis at Rehang Shackle Data
- Bird On Evis Verify 2 (if configured)
- Bird On Evis Verify 3 (if configured)
- Bird On Evis At End

**Note:** Tabs will only be available for verify brackets configured for your system.

## MAINTENANCE AND TROUBLESHOOTING

### DEBUG TOOLS

Each Debug Screen uses character codes to show the system status of a bird at that shackle. Red character codes indicate that the paws are being rejected. Black character codes mean the paws are not being rejected.

| Bird Code                                                                                      | Bird Location | Reject Status |                                                                                         |
|------------------------------------------------------------------------------------------------|---------------|---------------|-----------------------------------------------------------------------------------------|
| <b>k</b>                                                                                       |               |               | Bird was present on Kill Shackle, this code is only used on the All Shackle data report |
|                                                                                                | <b>1</b>      |               | Bird was present on Evis Verify At Start Shackle                                        |
|                                                                                                | <b>2</b>      |               | Bird was present on Evis Verify 2 Shackle                                               |
|                                                                                                | <b>3</b>      |               | Bird was present on Evis Verify 3 Shackle                                               |
|                                                                                                | <b>4</b>      |               | Bird was present on Evis Verify At End Shackle                                          |
|                                                                                                | <b>o</b>      |               | No Bird on Shackle                                                                      |
|                                                                                                | <b>u</b>      |               | No Bird on Shackle and Shackle Used to Reconcile Empty Shackle                          |
|                                                                                                |               | <b>g</b>      | Good, Paws Not Being Rejected                                                           |
|                                                                                                |               | <b>b</b>      | Paws Being Rejected Because of Bird Reject Button Press                                 |
|                                                                                                |               | <b>e</b>      | Paws Being Rejected Because Evis Line Behind or Stopped                                 |
|                                                                                                |               | <b>m</b>      | Paws Being Rejected Because Bird Missing                                                |
|                                                                                                |               | <b>a</b>      | Paws Being Rejected Because Reject All Turned On                                        |
| Special Codes used when Evis is running faster than Picking and Evis has extra empty shackles. |               | <b>xxx</b>    | Extra Evis Shackle Without Bird                                                         |
|                                                                                                |               | <b>xbx</b>    | Extra Evis Shackle With Bird                                                            |
|                                                                                                |               | <b>yyy</b>    | Extra Evis Shackle Without Bird and Shackle Used to Reconcile Empty Shackle             |
|                                                                                                |               | <b>yby</b>    | Extra Evis Shackle With Bird and Shackle Used to Reconcile Empty Shackle                |
|                                                                                                |               |               |                                                                                         |





Bird on Evis Verify at End





Bird On All

Bird on Evis Verify at End

|     |     |     |     |     |     |     |     |     |     |     |  |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|--|
| 14g | 14g | 14g | 14g |     |     |     |     |     |     |     |  |
| 14g | 14g | 14g | 14g |     |     |     |     |     |     |     |  |
| 14g | 14g | 14g | 14g |     |     |     |     |     |     |     |  |
| 14g | 14g | 14g | 14g | 14g | 14b | 14b | 14b | 14b | 14b | 14b |  |
| 14b | 14b | 14b | 14b | 14b | 14g | 14g | 14g | 14g | 14g | 14g |  |
| 14g | 14g | 14g | 14g | 14g |     |     |     |     |     |     |  |

System Console

Totals Report

Counters

Debug

**14b**

1 = bird present at evis verify 1  
4 = bird present at evis verify 4 (at end)  
b = paws rejected because reject button was pressed

The Bird Reject Before/After setting is 5 before and 5 after, so 1 bird reject button press caused 11 paw pairs to be rejected.

## No Bird Lights

Some locations install **No Bird Lights** near the Kill and Evis Line photoeyes. These lights help verify that the photoeyes are seeing birds properly as they travel past the photoeyes.

## Correct Functionality

- If a bird is not seen by the photoeyes, the No Bird Light will turn on for half a second.
- If no birds are detected on multiple shackles in a row, the No Bird Light will stay on until a shackle with a bird is detected.
- The No Bird Light should never be on when a loaded shackle passes by the sensors.

## Troubleshooting

**The No Bird Light comes on when a bird is on a shackle.**

Possible Causes:

- Sensor may be dirty.
- Sensor may not be aligned properly.
- Sensor has failed completely.
- Birds are swinging too much as they pass the sensors.

**Note:** This would normally only occur if the sensors are mounted too close to a point on the line where the birds make a sharp turn or when they are exiting another piece of equipment.



## MAINTENANCE AND TROUBLESHOOTING

### TEST BIRD MISSING PAW REJECTION

#### 2. Check the data for a red **KO**.

KO means a bird was found missing and the paws have been tagged for rejection when they reach the paw reject gate.

There may also be red **KE** shackles on both sides of the red **KO** value if the bird missing before and bird missing after values as specified on the System Setup 4 are greater than 0.

#### 3. Watch the kill line right before the rehang and the evis line right after the rehang.

##### Watch the Kill Line for:

- No empty shackles on the kill line
- AND an evis shackle without a bird

**If both occurred**, a bird must have fallen off in the rehang or possibly the evis line is running faster than the kill line resulting in empty shackles being pushed onto the evis line.

If the lines are running at the same speed and an empty evis shackle is seen, most likely a bird fell off in the rehang and paws should have been tagged for rejection. Immediately go the **Bird On Kill and Evis Shackle Data** debug screen.

##### Check the data for a red **KO**

**KO** means the system detected a missing bird and flagged the paws for rejection.

There may also be red **KE** shackles on both sides the red **KO** value if the bird missing before and bird missing after values are greater than 0.

**If you do not see a red **KO**** verify that the first value to the right of Search Shackles on System Setup 4 is not set to 500, which disables paw rejection when birds are missing.

#### 4. If using **No Bird Lights**, verify the evis line sensors are working properly by removing the piece of black tape from the evis line bird sensors and verifying that the No Bird Light comes on and stays on until putting the tape back over the photoeyes.



## MAINTENANCE AND TROUBLESHOOTING

### TEST BIRD MISSING PAW REJECTION

1. **Check the data for a red EO.**

EO means a bird was found missing and the paws have been tagged for rejection when they reach the paw reject gate.

There may also be red EA shackles on both sides of the red EO value if the bird missing before and bird missing after values as specified on the System Setup 4 are greater than 0.

2. **Watch the evis line** at the sensors right after the rehang and at the sensors shortly before the first USDA button.

**Watch the Evis line for:**

- No empty shackles at the sensors after the rehang or along the evis line
- AND an evis shackle without a bird at the sensors before the Evis At End Photoeyes.

**If both occurred**, a bird must have fallen off along the evis line and paws should have been tagged for rejection. Immediately go to the **Bird On Evis and Adjust Before Shackle Data** debug screen.

**Check the data for a red EO**

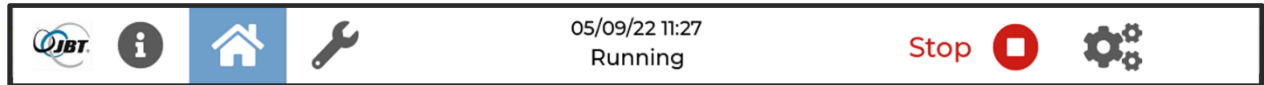
EO means a bird was found missing and the paws have been tagged for rejection when they reach the paw reject gate.

There may also be red EA shackles on both sides of the red EO value if the bird missing before and bird missing after values as specified on the System Setup 4 are greater than 0.

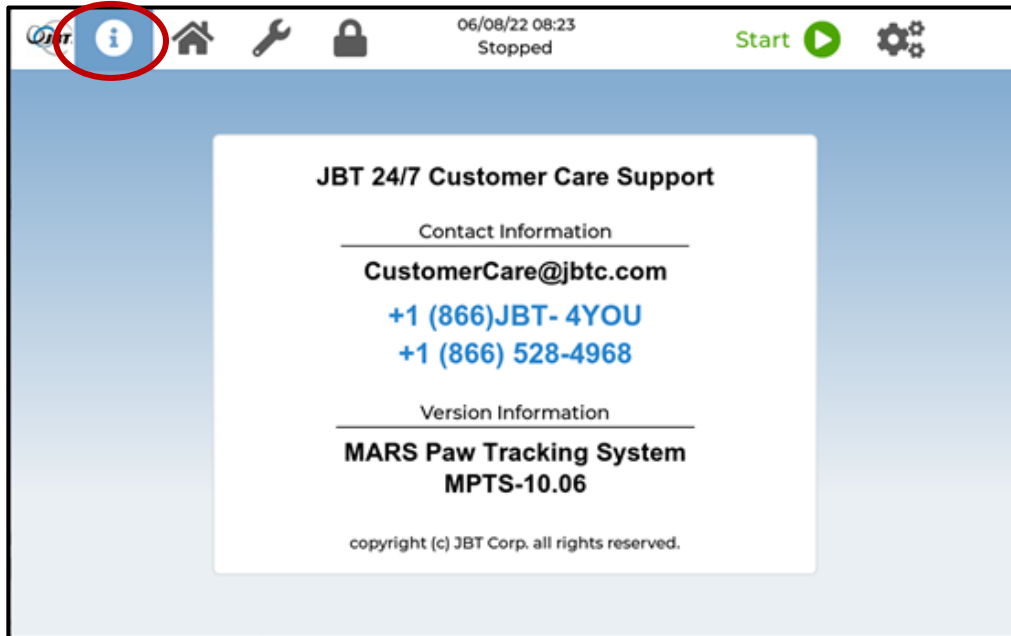
**If you do not see a red EO** verify that second value to the right of Search Shackles on System Setup 4 is not set to 500 which disables paw rejection when birds are missing.

3. If using **No Bird Lights**, verify the evis line sensors are working properly by removing the piece of black tape from the evis line bird sensors and verifying that the No Bird Light comes on and stays on until putting the tape back over the photoeyes.

### Info and About



#### Company Contact and Software Version Information



### Contact Info

Phone: 1 - 866 – JBT – 4YOU  
1 - 866 – 528 - 4968  
Email: CustomerCare@jbtc.com

### Contact a JBT Service Technician

If you have a question about your **JBT System** or require service after trying to troubleshoot the issue, please contact us using the contact information provided on your system's Info screen.

### Troubleshooting

The **Hardware Test** screen is a good place to start when experiencing problems with the correct operation of your **JBT System**.

If all the external hardware appears to be functioning properly on the **Hardware Test** screen, then diagnosing the problem may require assistance from a JBT service technician.



## ASSEMBLY DRAWINGS

The following pages include the Bill of Materials and Assembly Drawings for

## CONTROLS AND ELECTRICAL SCHEMATICS

| PART NO. | DESCRIPTION |
|----------|-------------|
|          |             |
|          |             |

## SPARE PARTS LIST

To order spares call 866.528.4968

Email: [parts.prime@JBTC.com](mailto:parts.prime@JBTC.com)

Refer to serial number:

Refer to model number: JBT GEN2 Paw Tracking System

Although many parts may seem similar between 2 machines, due to wear and adjustments they become machine specific. You will cause damage to your machine if you swap parts with other machines.

| PART NO. | DESCRIPTION                                  | QTY. |
|----------|----------------------------------------------|------|
| 11825    | PWRSPLY,100-550VAC,24VDC*                    | 2    |
| 80001    | BOARD, CNTRL, JBT                            | 1    |
| 80036    | TERMINAL, HLDR, FUSE, MINI, 24V              | 5    |
| 80037    | FUSE, 2A, MINI                               | 10   |
| 80038    | FUSE, 5A, MINI                               | 5    |
| 80742    | DISPLAY, 10", PNL MT, ALUM BEZEL, IP65       | 1    |
| 80787    | CBL, HDMI, 3FT                               | 3    |
| 81370    | SOL,5/2 WAY,SUB-BASE,24V DC                  | 1    |
| MRS1241  | LIGHT,30MM, 10-30V,G-R-Y,2M CABLE            | 4    |
| MRS1250  | BOX ASSY,COUNTER,EVIS                        | 1    |
| MRS1276  | Tsunami Filter-Regulator Combo               | 2    |
| MRS1284  | Illuminated Push Button Assembly – Momentary | 6    |
| MRS1286  | Illuminated Maint Button Assy (Committed)    | 2    |
| MRS6001  | Hardwire Send Sensor w/30' Cable             | 12   |
| MRS6011  | Hardwire Receive Sensor 2/9M' Cable          | 12   |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |
|          |                                              |      |

## TECHNICAL PARTS INFORMATION

| PART NO. | DESCRIPTION |
|----------|-------------|
|          |             |
|          |             |